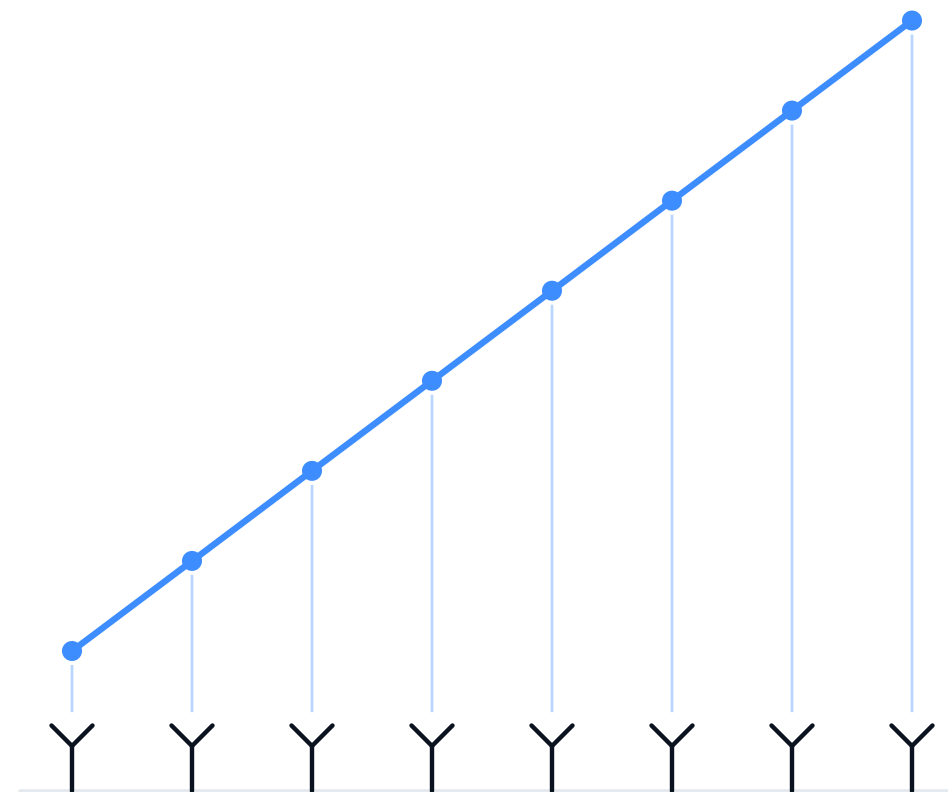


MIMO

One array, four uses

An antenna array turns a direction into a vector of phases. Adding the copies up is diversity; making two of them differ is spatial multiplexing; choosing them is beamforming; reading them is how a radio finds where you are. This lecture is one object, asked four questions.

Yaxiong Xie



$a(\theta)$ · the phase ramp

Why do we need MIMO at all?

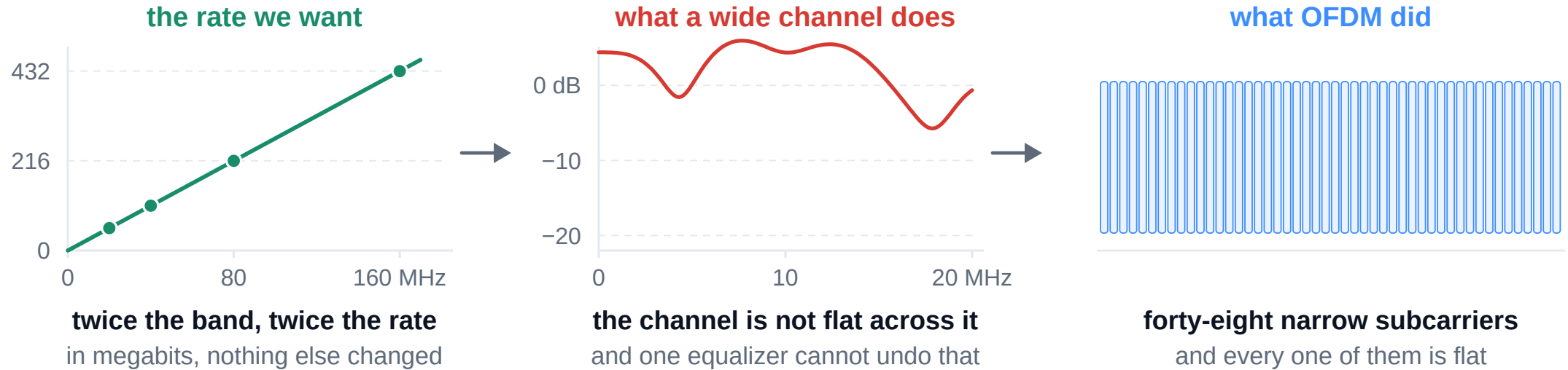
01

Why MIMO

Bandwidth is what the last lecture bought us, and OFDM is what made it usable. This part asks what is left after that: how strong the signal is when it arrives, why the transmitter cannot simply shout louder, and what is still unspent.

OFDM is what let us use a wide channel

WHY WE WANTED A WIDE CHANNEL, AND WHAT IT TOOK TO GET ONE



Bandwidth is the one term in the rate that pays back one for one, and OFDM is what made it spendable.

802.11a: 20 MHz of channel, 48 usable subcarriers, 54 Mbit/s.

That was the last lecture: bandwidth, made usable.

1 Why MIMO

2 Diversity

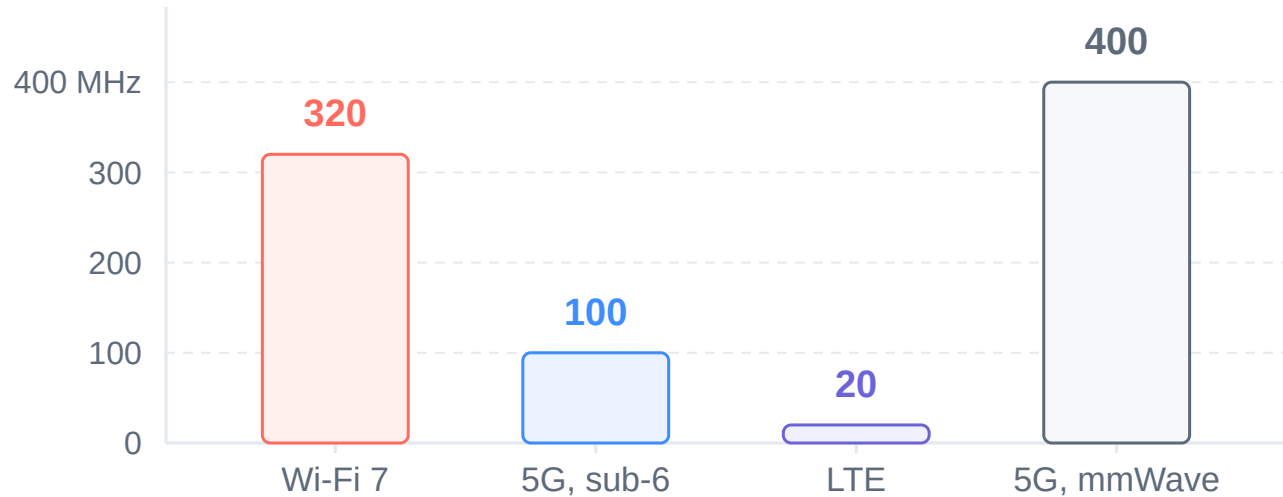
3 Multiplexing

4 Beamforming

5 Together

A channel is a few hundred megahertz, licensed or not

THE WIDEST CHANNEL EACH STANDARD DEFINES



Licensed or unlicensed, the number comes out the same size.

The grey bar is above 6 GHz, where the range collapses. That is Part 4.

unlicensed

about 2,000 MHz across 2.4, 5 and 6 GHz where the 6 GHz band is open, and shared

licensed

a block bought at auction, and the same order of magnitude for one operator

either way

one link ends up with a few hundred megahertz, and the standards already use it

Widest channel each standard defines. Carrier aggregation stacks several of them together, and pays for the spectrum under every one.

Bandwidth works, and no system is going to get much more of it.

1 Why MIMO

2 Diversity

3 Multiplexing

4 Beamforming

5 Together

So what else is there, besides bandwidth?

THE RATE IS A PRODUCT OF TWO THINGS

$$\text{rate} = \text{bandwidth} \times \text{bits per second per hertz}$$

bandwidth

OFDM made it usable
and the regulator says how much there is

✓ already spent

OFDM used it, and 2,000 MHz is all there is

bits per second per hertz

?



how strong the signal is

when it arrives at the receiver
and that is the second axis

The rest of Part 1 is what that second axis costs.

The other way to go faster is to put more signal on the receiver.

1 Why MIMO

2 Diversity

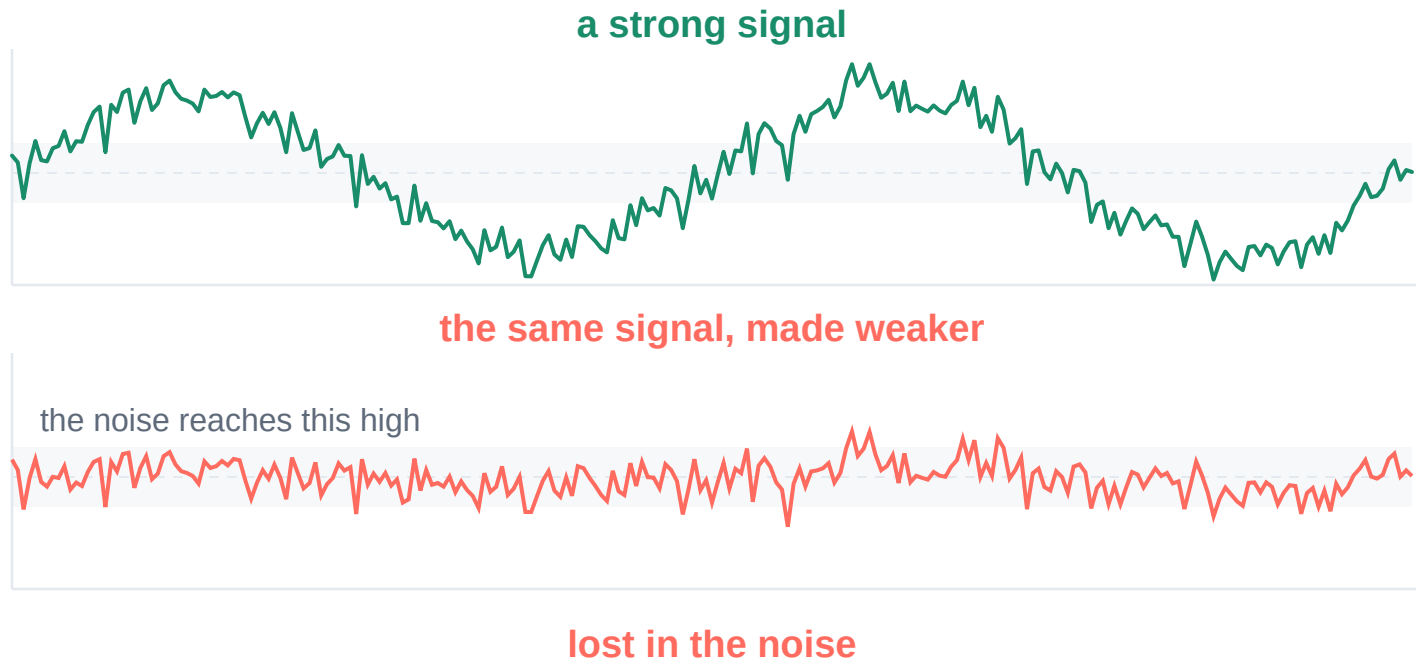
3 Multiplexing

4 Beamforming

5 Together

A weak signal disappears into the noise

THE SAME NOISE, A STRONG SIGNAL AND A WEAK ONE



the noise

the same size in both, because it comes from temperature

the signal

the only thing the transmitter gets to change

$$\text{signal} \div \text{noise} = \text{SNR}$$

Signal-to-noise ratio is the name for how far the signal sticks out of the noise.

1 Why MIMO

2 Diversity

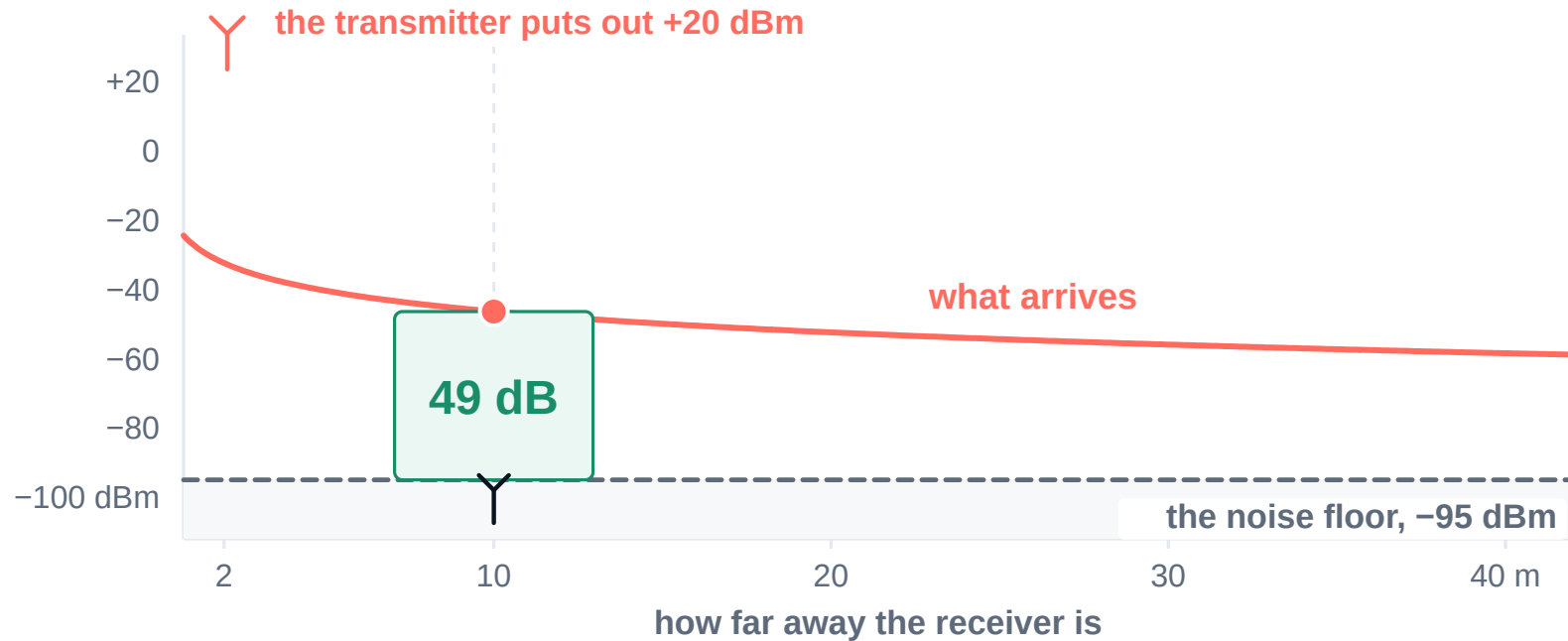
3 Multiplexing

4 Beamforming

5 Together

Most of the signal is lost, and the noise floor stays put

ONE LINK, 5 GHz, IN CLEAR VIEW



the signal
falls off as the receiver moves away

the noise floor
does not move at all:
it is set by temperature
and by the bandwidth

Free space at 5 GHz, one antenna at each end. Walls and furniture push the curve down; they do not move the floor.

Signal-to-noise ratio is the gap between those two lines, and only the top one moves.

1 Why MIMO

2 Diversity

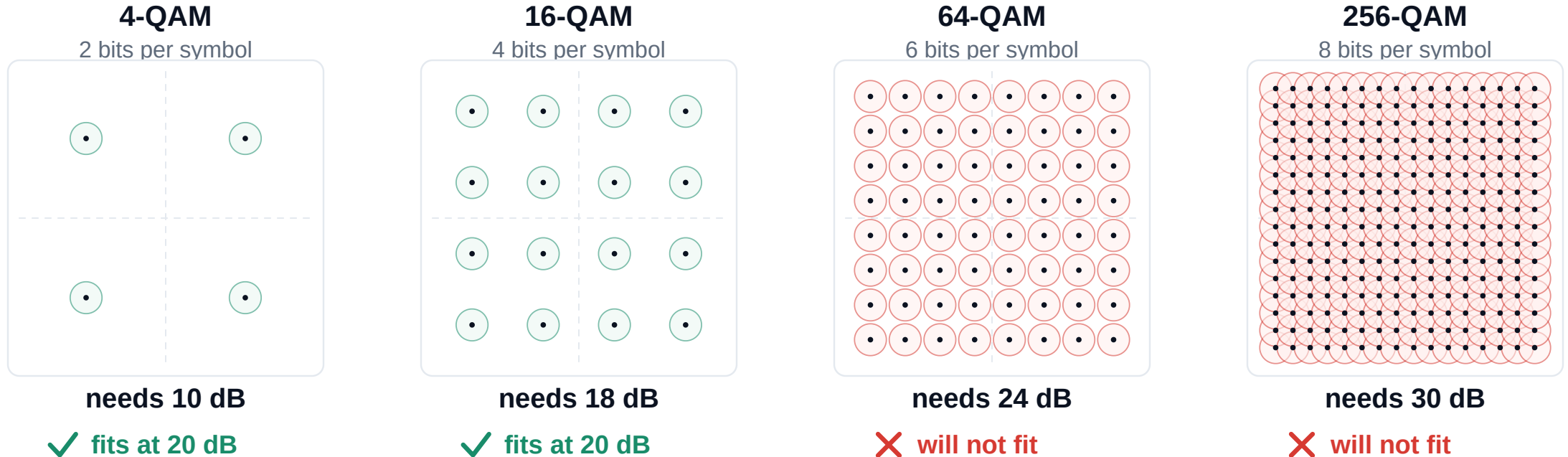
3 Multiplexing

4 Beamforming

5 Together

The gap decides how many points fit in one symbol

THE SAME NOISE, AND FOUR WAYS TO DIVIDE UP ONE SYMBOL



Every circle is the same 20 dB of noise. The constellations all carry the same average power, so more bits means the points move closer together.

Bits per symbol is a question about how much room the noise leaves.

1 Why MIMO

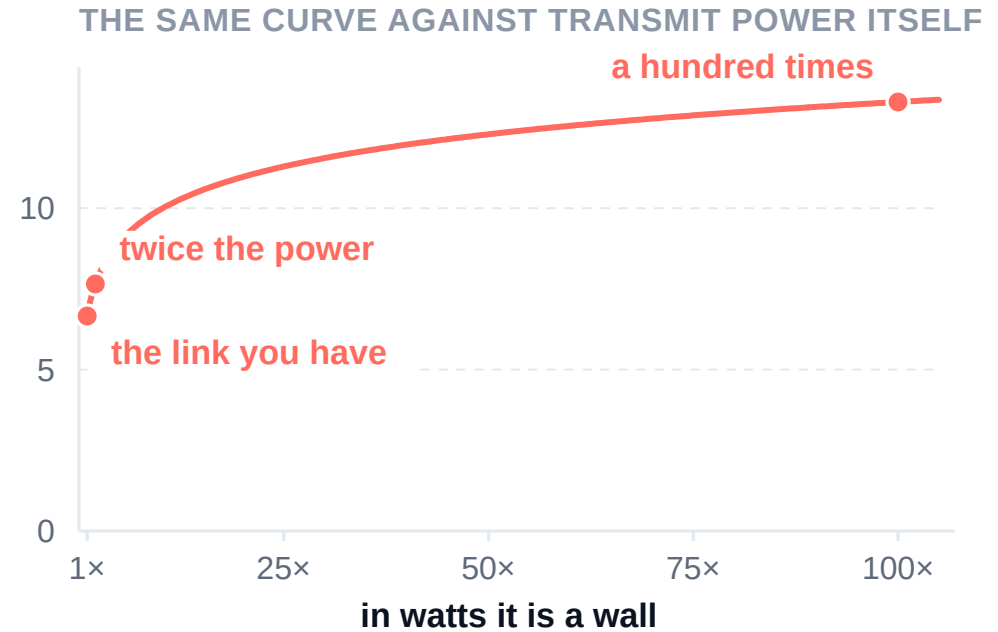
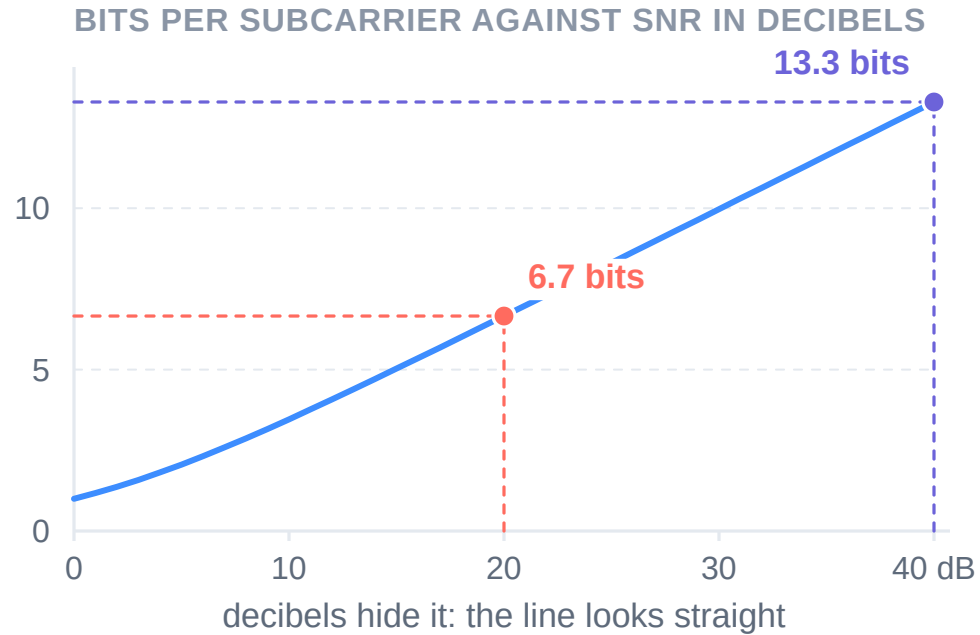
2 Diversity

3 Multiplexing

4 Beamforming

5 Together

Every extra bit costs twice the transmit power



Twice the power is one more bit per subcarrier. A hundred times the power doubles the rate.

The second knob is a logarithm, so pushing it hard buys very little.

1 Why MIMO

2 Diversity

3 Multiplexing

4 Beamforming

5 Together

The transmitter cannot simply turn the power up

AND TWICE THE POWER IS NOT AVAILABLE EITHER

the regulator

unlicensed transmit power is capped at a fraction of a watt, and the cap differs by band and region

the amplifier

1024-QAM needs a linear amplifier, so it already runs backed off from its peak

the battery

the transmit chain is one of the largest draws in a phone, and it is not linear either

the distance

twice the power is 3 dB, which is 1.4 times the range or one more bit — not both

the other end

the uplink has the same cap, a smaller battery and a worse antenna

Both branches are now closed, and every radio so far has used exactly one point in space.

More power is the obvious answer, and it is the one that ran out first.

1 Why MIMO

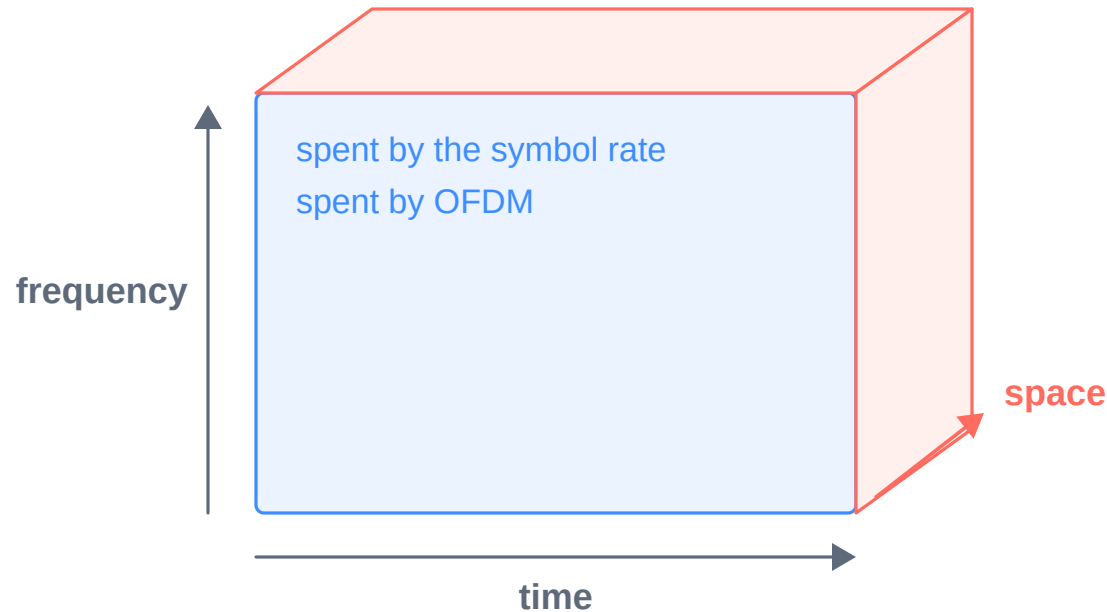
2 Diversity

3 Multiplexing

4 Beamforming

5 Together

Everything so far left one antenna at each end



axis	what spends it	left
time	the symbol rate	none
frequency	the channel width	none
space	—	all of it

One antenna at each end uses one point in space.
Every design so far has done exactly that.

Time and frequency are both fully spent. The third axis has not been touched by anything in the OFDM lecture.

Space is the one axis with room left in it.

1 Why MIMO

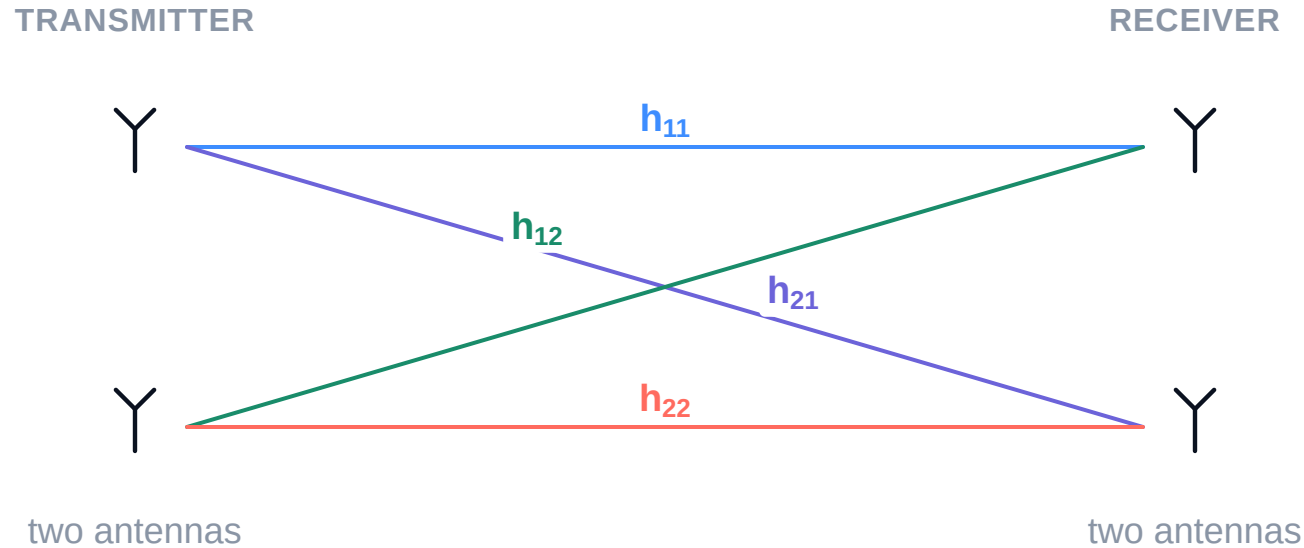
2 Diversity

3 Multiplexing

4 Beamforming

5 Together

Two antennas at each end make four channels



two antennas at each end give four separate channels

Every transmit antenna reaches every receive antenna, so the link stops being one number.

One antenna pair gave one number. Two pairs give a 2×2 matrix.

1 Why MIMO

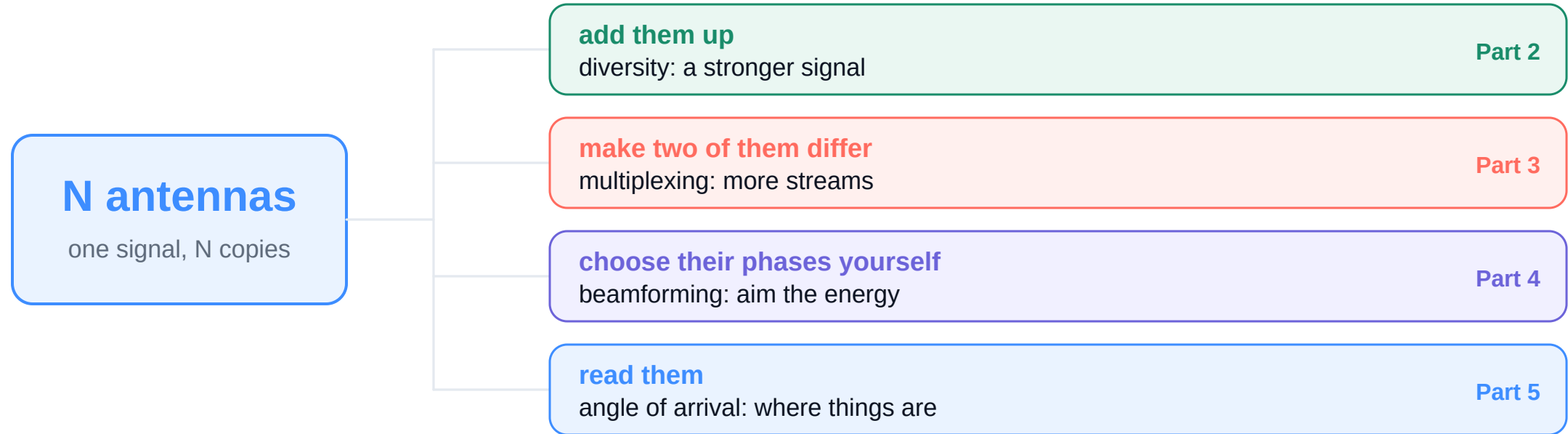
2 Diversity

3 Multiplexing

4 Beamforming

5 Together

Four things to do with a second antenna



Each of the next four parts asks one question of the same set of copies, and answers it with the same weighted sum.

Everything after this is arithmetic on the copies the room delivers.

1 Why MIMO

2 Diversity

3 Multiplexing

4 Beamforming

5 Together

What is the simplest thing to do with the copies?

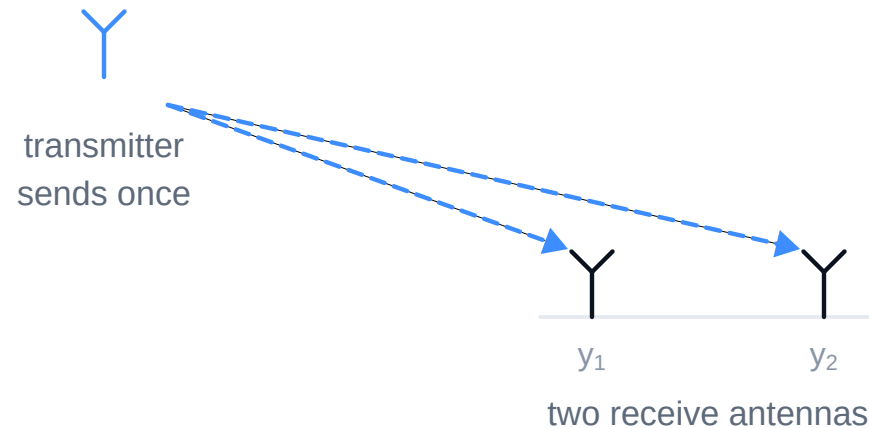
02

Diversity

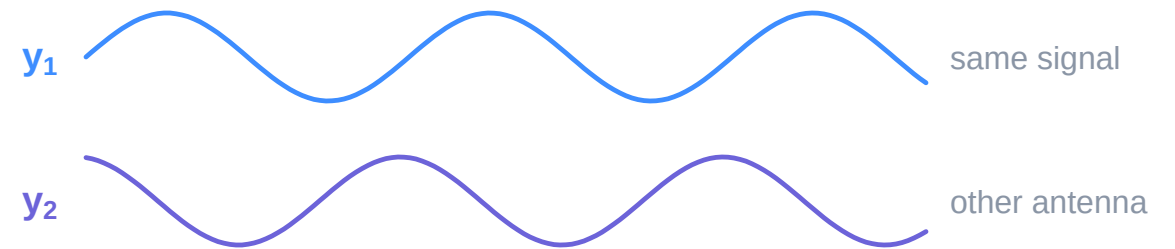
Two antennas mean two copies of the same signal, and the simplest thing to do with them is add them up. Whether that works depends on the phase each one arrives with, so this part measures the phase across the array and then spends it.

Two antennas, one signal: can we just add them?

ONE TRANSMIT ANTENNA, TWO RECEIVE ANTENNAS



WHAT ARRIVES AT EACH ONE



$$y_1 + y_2 = ?$$

two copies of one signal — why not just add them up?

The two antennas are half a wavelength apart and the signal is the same one, sent once. Nothing is decided on this slide.

Adding the two antennas is the obvious move. Does it work?

1 Why MIMO

2 Diversity

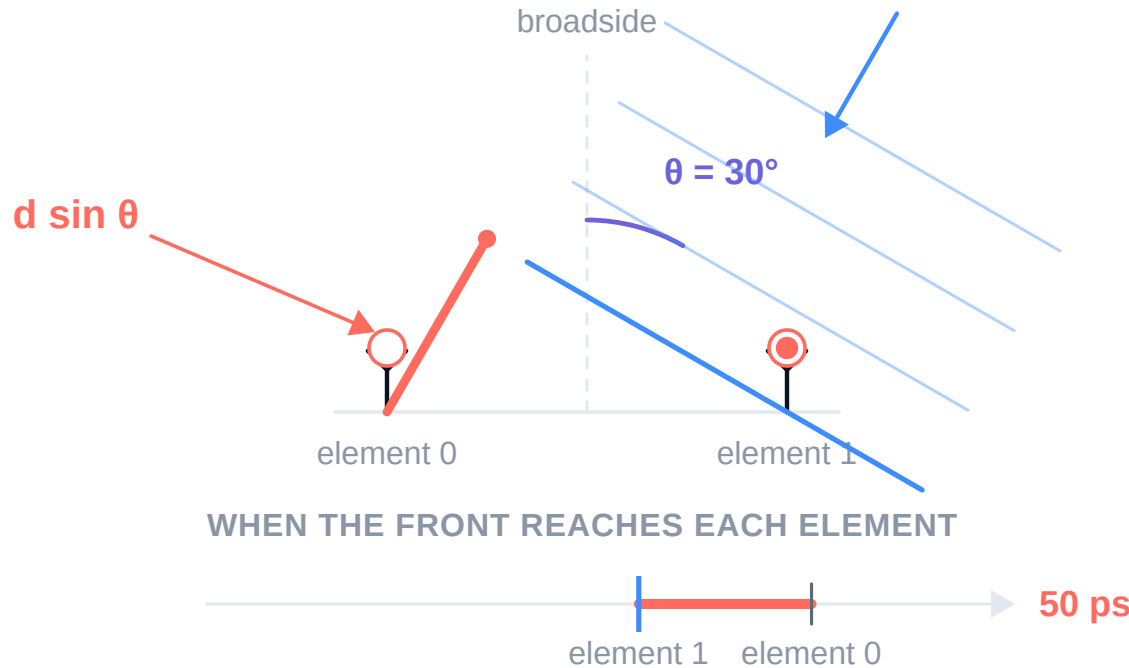
3 Multiplexing

4 Beamforming

5 Together

A wave reaches two antennas at two different times

A PLANE WAVE ARRIVING FROM 30° OFF BROADSIDE



WHEN THE FRONT REACHES EACH ELEMENT

θ is measured from broadside. The source is far enough away that the arriving fronts are straight lines.

ONE WORKED CASE

carrier	5 GHz
λ	6.0 cm
$d = \lambda/2$	3.0 cm
θ	30°
extra path $d \sin \theta$	1.5 cm
extra time	50 ps

The front touches element 1 first and travels $d \sin \theta$ further to reach element 0.

1 Why MIMO

2 Diversity

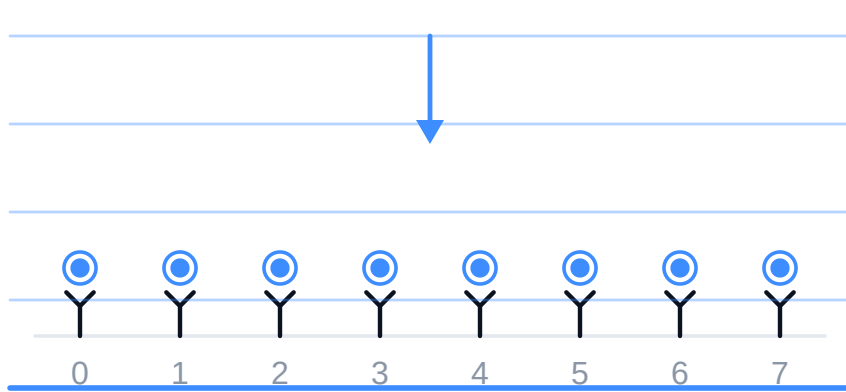
3 Multiplexing

4 Beamforming

5 Together

Straight on, all eight antennas see the same phase

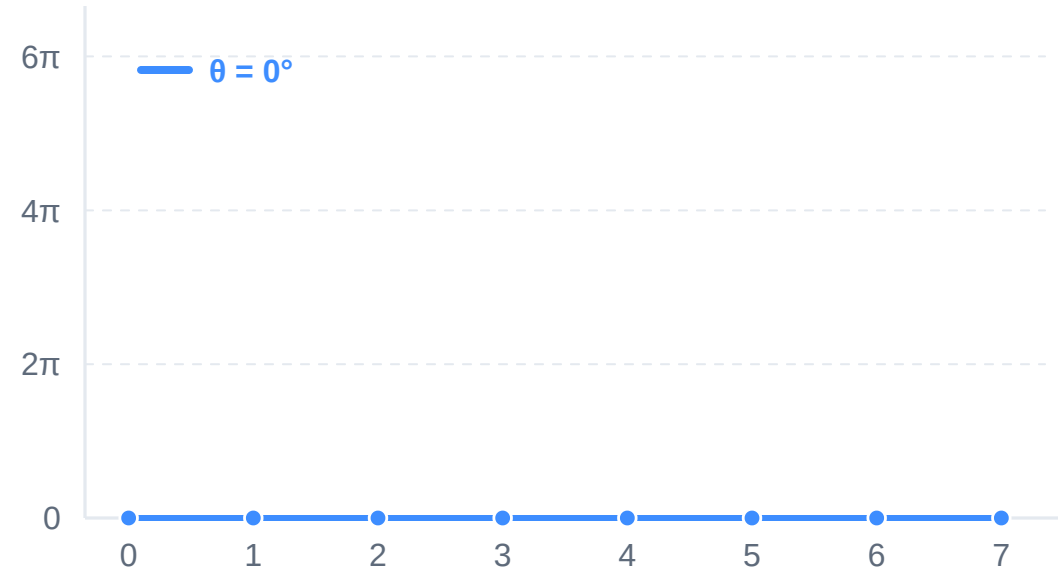
EIGHT ELEMENTS, HALF A WAVELENGTH APART



the front touches all eight at the same instant

$\theta = 0^\circ$

PHASE OF ELEMENT n



$$0^\circ = 180^\circ \times \sin 0^\circ$$

θ is measured from broadside, so a wave arriving perpendicular to the array is $\theta = 0^\circ$. Drawn unwrapped so the slope is visible.

A flat ramp is what broadside looks like.

1 Why MIMO

2 Diversity

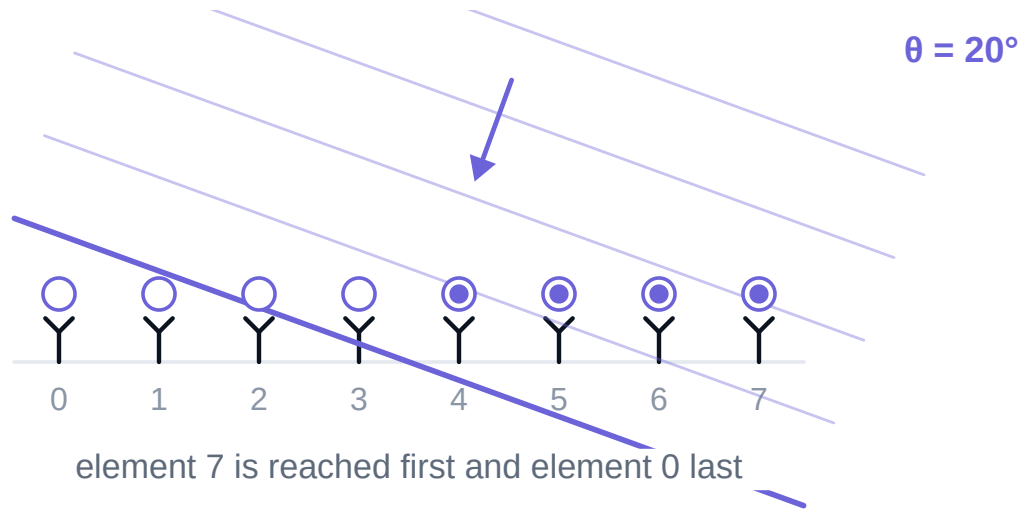
3 Multiplexing

4 Beamforming

5 Together

Tilt the wave 20° and every antenna is one step behind

EIGHT ELEMENTS, HALF A WAVELENGTH APART

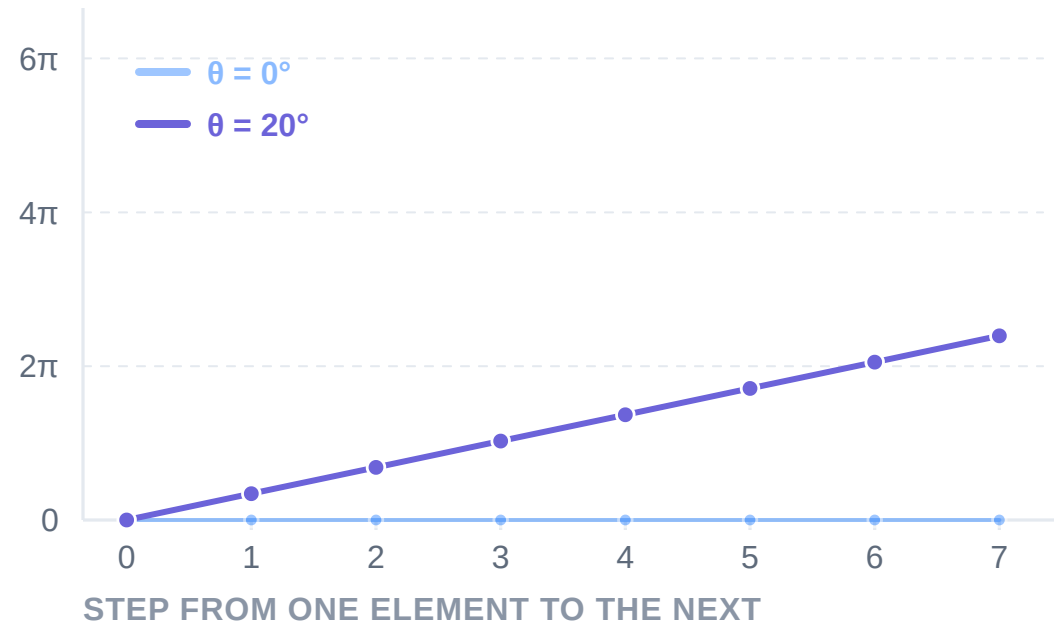


element n is $n \cdot \pi \sin \theta$ behind element 0

$$a(\theta)_n = \exp(j \cdot n \cdot \pi \cdot \sin \theta)$$

the N phases together are the steering vector

PHASE OF ELEMENT n



$$61.6^\circ = 180^\circ \times \sin 20^\circ$$

Drawn unwrapped so the slope is visible; a receiver measures each phase modulo 2π . The 0° line from the last slide is kept for comparison.

The angle sets the slope of the ramp, and nothing else does.

1 Why MIMO

2 Diversity

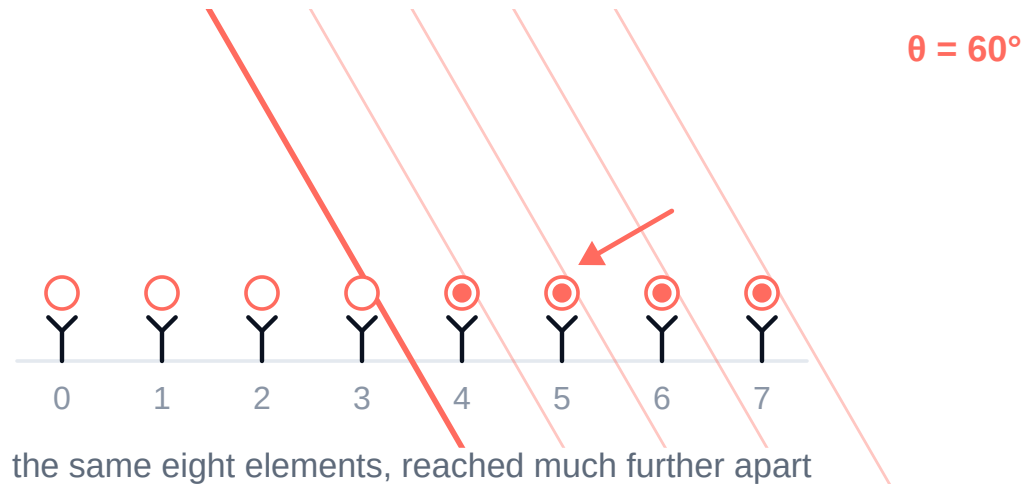
3 Multiplexing

4 Beamforming

5 Together

At 60° the step is 156° and the ramp is steep

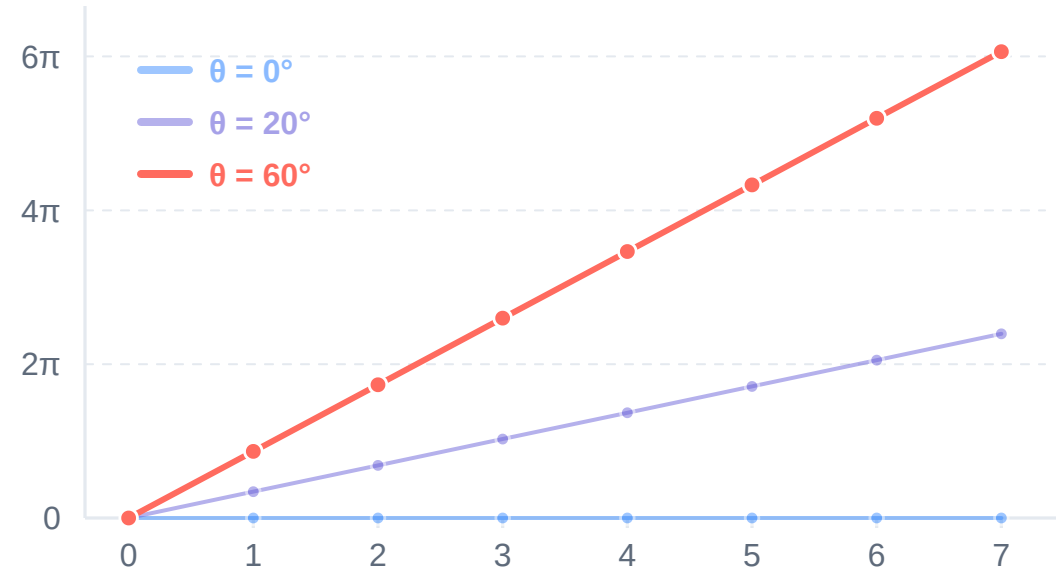
EIGHT ELEMENTS, HALF A WAVELENGTH APART



three times the angle is not three times the step
 $\sin \theta$ is what counts, and at 90° the step stops at 180°
 which is as far as half-wavelength spacing can go

All three angles on one axis. Drawn unwrapped; a receiver measures each phase modulo 2π , so it sees these ramps folded into $\pm 180^\circ$.

PHASE OF ELEMENT n



STEP FROM ONE ELEMENT TO THE NEXT

$$155.9^\circ = 180^\circ \times \sin 60^\circ$$

Eight phases, one number: the slope says where the wave came from.

1 Why MIMO

2 Diversity

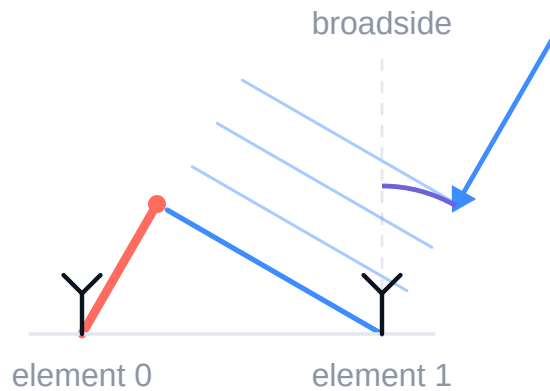
3 Multiplexing

4 Beamforming

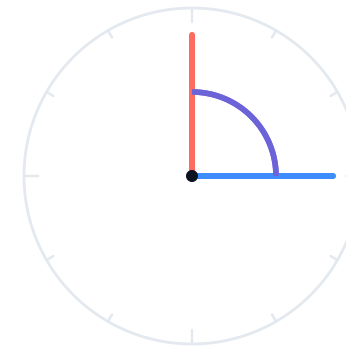
5 Together

An extra $d \sin \theta$ of path is a phase difference

A PLANE WAVE ARRIVING FROM θ



PHASE OF ELEMENT 1



— element 0, the reference
— element 1

θ
 30°

EXTRA PATH
 1.50 cm

OF A CYCLE
 0.250

PHASE $\Delta\phi$
 90°

$$\Delta\phi = 2\pi \cdot d \sin \theta / \lambda$$

$$d = \lambda/2 \rightarrow \Delta\phi = 180^\circ \times \sin \theta$$

The geometry is frozen and only the arrival angle moves; one sweep runs from -90° through broadside to $+90^\circ$.

A phase difference is all the array has, and the angle is what sets it.

1 Why MIMO

2 Diversity

3 Multiplexing

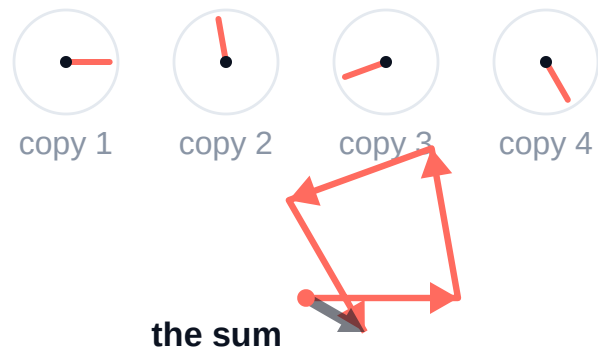
4 Beamforming

5 Together

Add the copies as they arrive and they cancel

FOUR COPIES OF ONE SIGNAL, ONE PER ANTENNA

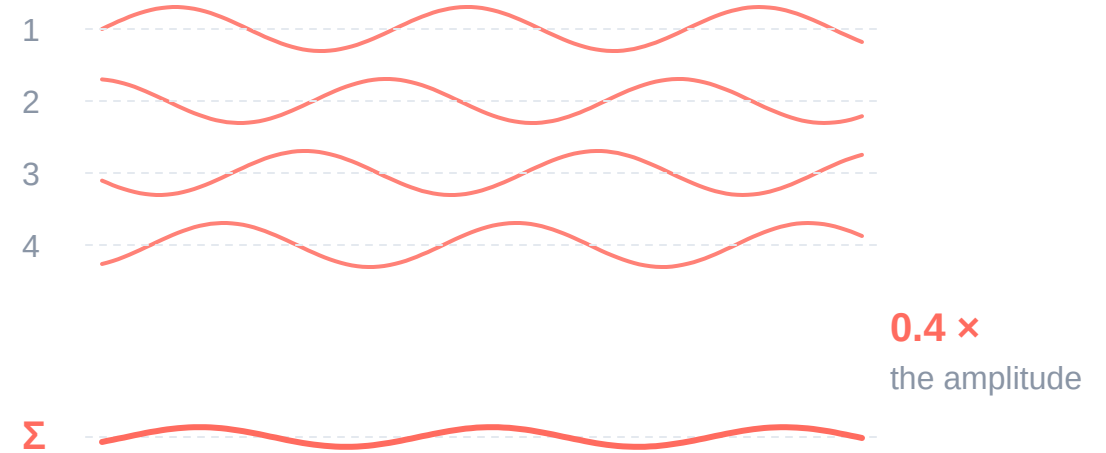
EACH COPY ARRIVES WITH ITS OWN PHASE



THE SUM, IN COPIES

0.4 of 4

THE SAME FOUR COPIES, AS WAVEFORMS



Four arrows of equal length, added head to tail, and the same four copies as waveforms. In the browser the phases drift the way a moving channel drifts and the sum swings between four copies and nothing.

Two antennas can add up to twice the signal, or to none of it. The phase decides which.

1 Why MIMO

2 Diversity

3 Multiplexing

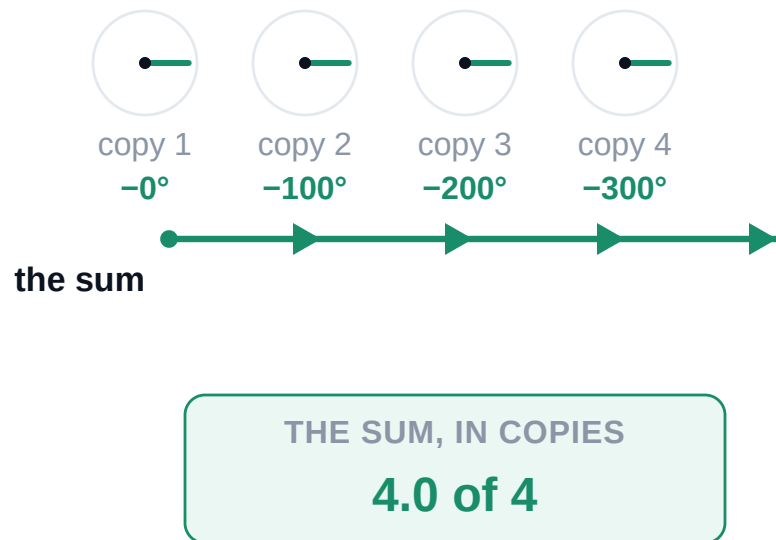
4 Beamforming

5 Together

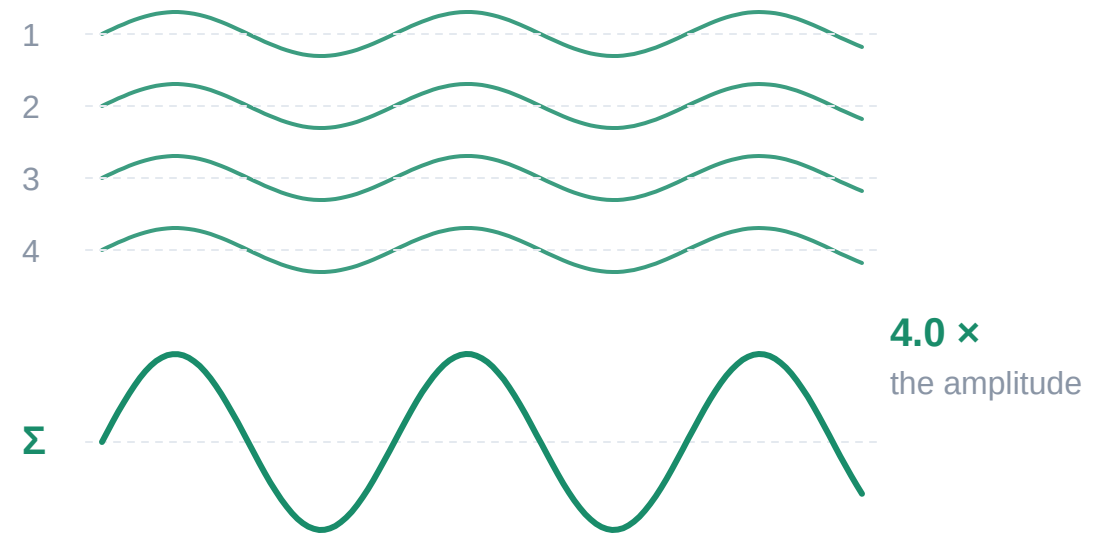
Turn every copy upright, then add

THE SAME FOUR COPIES, EACH ONE TURNED BACK FIRST

EVERY PHASE UNDONE, SO THEY ALL POINT ONE WAY



THE SAME FOUR COPIES, AS WAVEFORMS



The weight on antenna n is $e^{(-j\phi_n)}$: the conjugate of the phase that antenna measured. Nothing is amplified — the copies are only rotated into step.

Undo each phase, then add: four copies become four times the signal, every time.

1 Why MIMO

2 Diversity

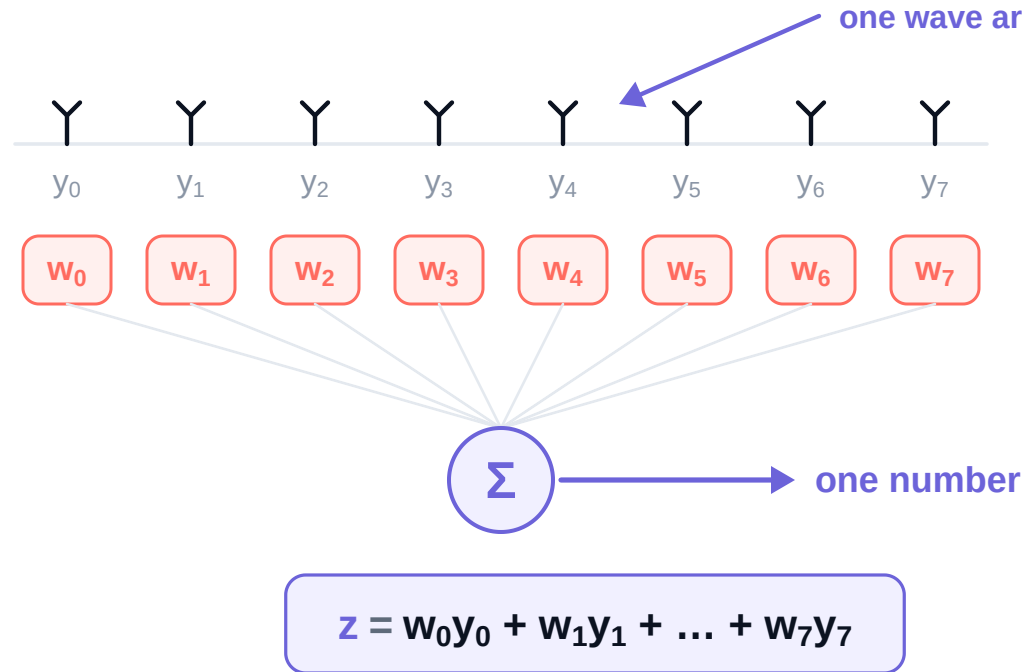
3 Multiplexing

4 Beamforming

5 Together

Everything an array does is one weighted sum

EIGHT ANTENNAS, EIGHT WEIGHTS, ONE NUMBER



The weights are complex, so each one rotates its antenna before the sum. The noise walk is one draw; on average eight of them add to $\sqrt{8}$.

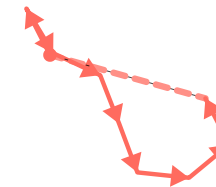
WHY ADD AT ALL

WEIGHTS MATCHED TO ONE DIRECTION

each weight turns its own sample to the same phase

→ → → → → → → → $8 \times \text{the amplitude}$

THE NOISE HAS NO DIRECTION TO MATCH



$\sqrt{8} \times \text{the amplitude}$

so the ratio $\times 8 = 9 \text{ dB}$

Beamforming, zero forcing and MMSE are this machine with three sets of weights.

1 Why MIMO

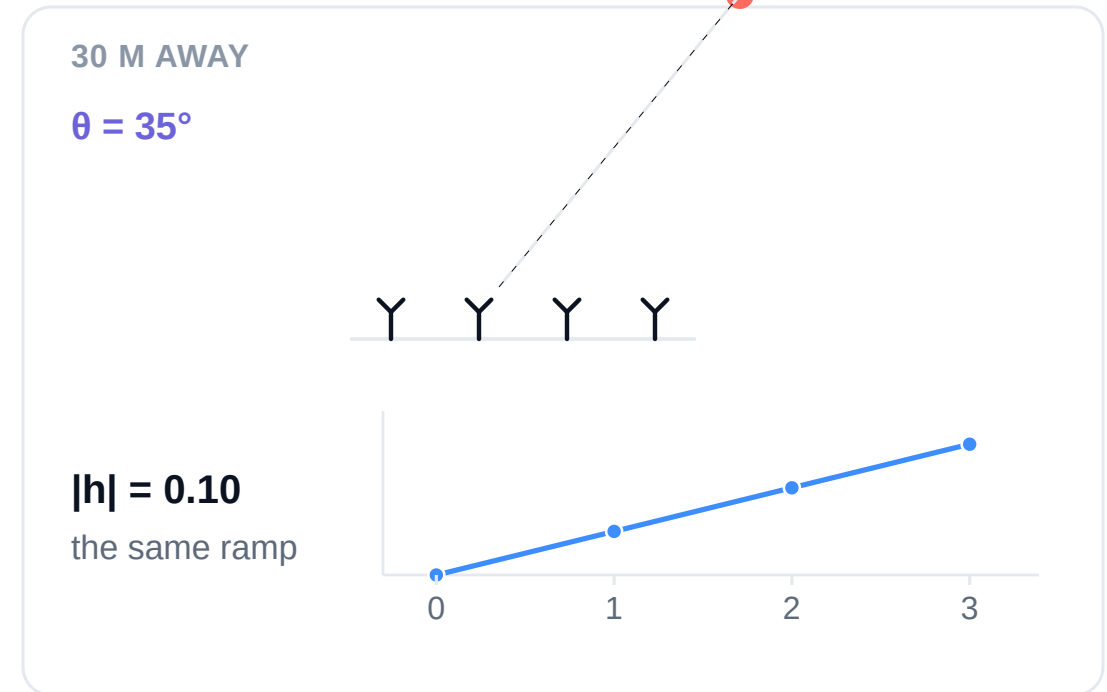
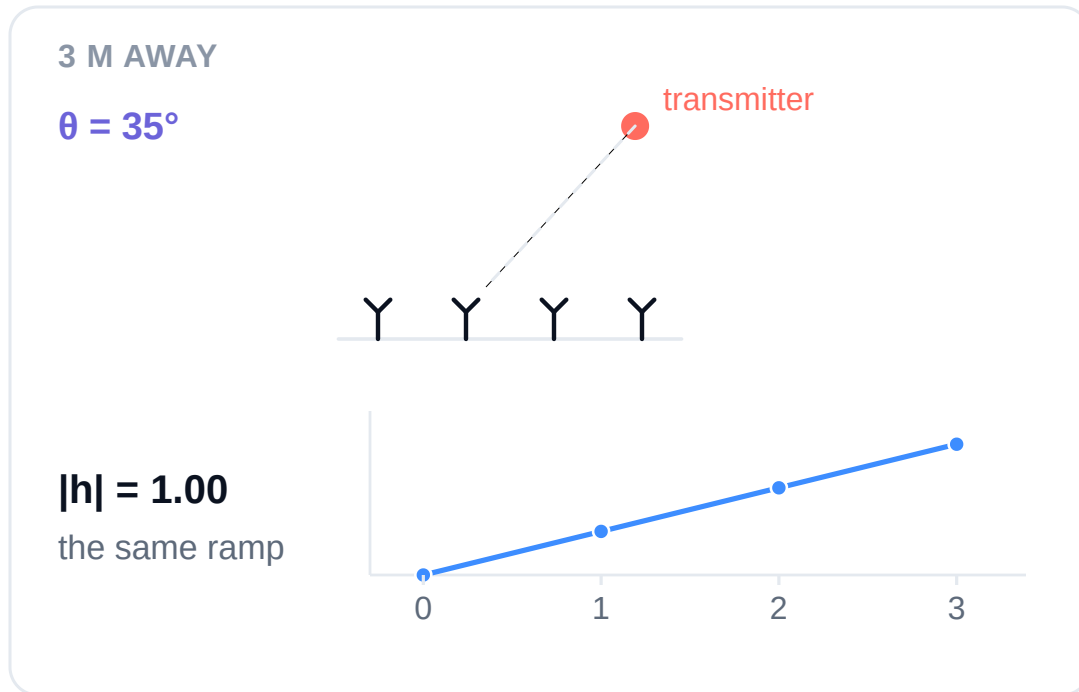
2 Diversity

3 Multiplexing

4 Beamforming

5 Together

The same ramp comes from 3 m away and from 30 m



Distance changes how loud the signal is. It does not change the phase difference between neighbouring elements.

An array measures direction. Range has to come from somewhere else.

1 Why MIMO

2 Diversity

3 Multiplexing

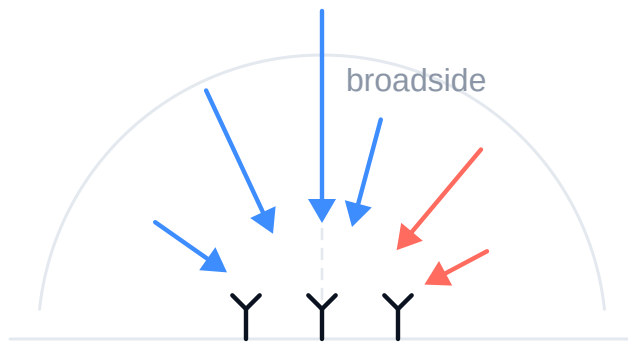
4 Beamforming

5 Together

A real channel is several directions added together

SIX PATHS, EACH ARRIVING FROM ITS OWN ANGLE

— positive gain — negative gain



one direction, two elements

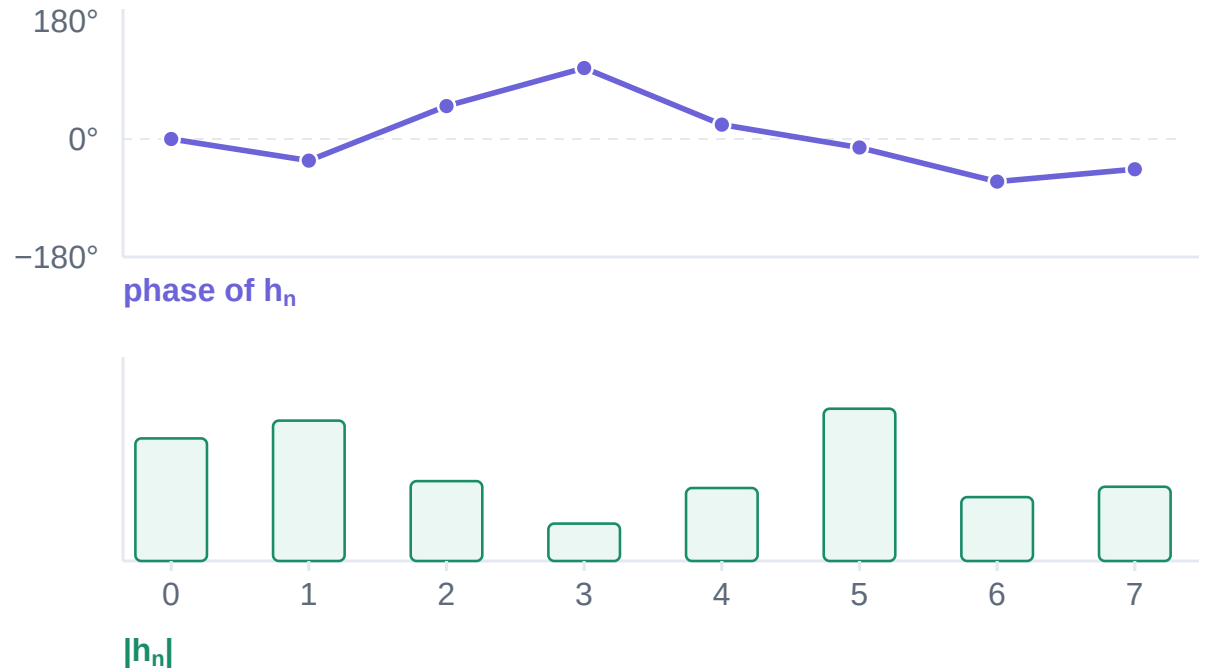
what differed was the time it reached each one

six directions, one array

what differs is the angle each one arrives from

Gains 1.00, 0.68, -0.52, 0.40, 0.26, -0.16 — the office channel from the OFDM lecture; the angles 0° , -25° , 40° , 15° , -55° , 62° are a choice here.

ONE COLUMN OF H , ELEMENT BY ELEMENT



One column of H is a sum of steering vectors, so it is no longer a straight ramp.

1 Why MIMO

2 Diversity

3 Multiplexing

4 Beamforming

5 Together

Maximal ratio combining, one step at a time

WHAT THE RECEIVER ALREADY MEASURED, AND WHAT IT DOES WITH IT



$$z = h_0^*y_0 + h_1^*y_1 + h_2^*y_2 + h_3^*y_3$$

$$\text{SNR out} = \text{SNR}_0 + \text{SNR}_1 + \text{SNR}_2 + \text{SNR}_3$$

The receiver already estimated h_n on every antenna, for its own equaliser. One multiply then does two jobs: the phase of h_n^* turns the copy back into step, and the size of it sets how loud that copy counts.

Multiply each antenna by the conjugate of its own channel, and the ratios add.

1 Why MIMO

2 Diversity

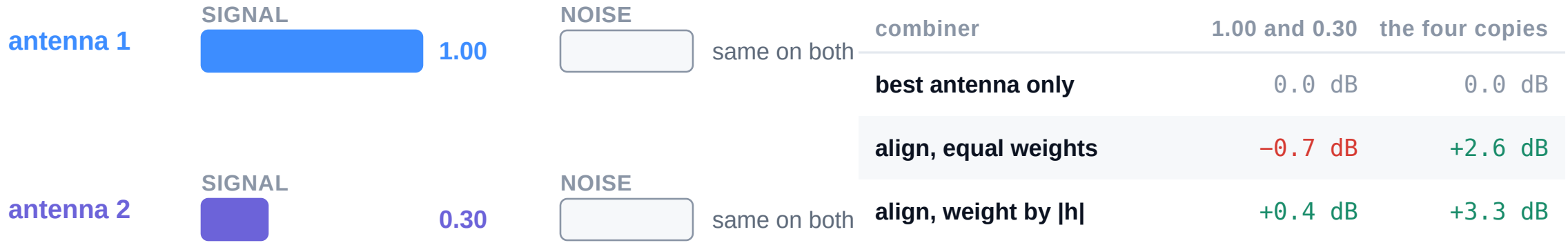
3 Multiplexing

4 Beamforming

5 Together

Why the weak antenna gets a smaller say

ONE STRONG ANTENNA AND ONE WEAK ONE



A third of the signal, and all of the noise.

Counted equally it drags the ratio down.

Both of the last two align the phase first.

They differ only in how loud each antenna is allowed to be.

MRC is never worse than the best antenna

Output signal-to-noise ratio against using the better antenna alone: $\max|h|^2$, $(\sum|h|^2)/N$ and $\sum|h|^2$, all three after co-phasing. With a very unequal pair, equal weights are worse than ignoring the weak one.

Every antenna gets a say in proportion to its own ratio. That is maximal ratio.

1 Why MIMO

2 Diversity

3 Multiplexing

4 Beamforming

5 Together

One antenna lands in a fade; the other does not

RECEIVED POWER ON TWO ANTENNAS, SIDE BY SIDE IN TIME



Ten scatterers with their own Doppler on each antenna, from a seeded generator, so the envelope is Rayleigh. The two percentages are exact: $1 - e^{(-x)}$ against $1 - e^{(-x)}(1 + x)$ at $x = 0.2$.

Diversity does nothing for the average and everything for the worst moments.

1 Why MIMO

2 Diversity

3 Multiplexing

4 Beamforming

5 Together

Diversity is already in everything you own

WHERE THIS IS ALREADY RUNNING

every access point

2 to 8 antennas

it combines them even for a client that sends one stream



every phone

2 to 4 antennas

receive combining is free: no feedback, no standard support



LTE and NR

two branches, minimum

receive diversity is in the specification, not an option



None of this asks the transmitter to know anything.

Every antenna a receiver has, it can add in — and the gain lands in the packets that used to fail.

Receive combining asks nothing of the other end — no feedback, no cooperation, no standard support — which is why it is in everything.

The first thing anyone does with spare antennas is add them together.

1 Why MIMO

2 Diversity

3 Multiplexing

4 Beamforming

5 Together

How do we send more than one signal at once?

03

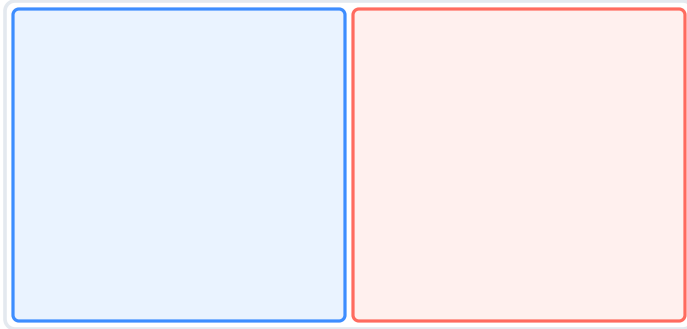
Multiplexing

Adding the copies up buys signal strength, and signal strength buys bits slowly. So far both transmit antennas sent the same signal. This part sends two different ones at the same instant on the same subcarrier: first what the room has to look like for that to be possible, and then the matrix that does it.

The two signals overlap in time and in frequency

THREE WAYS TWO SIGNALS CAN SHARE A CHANNEL

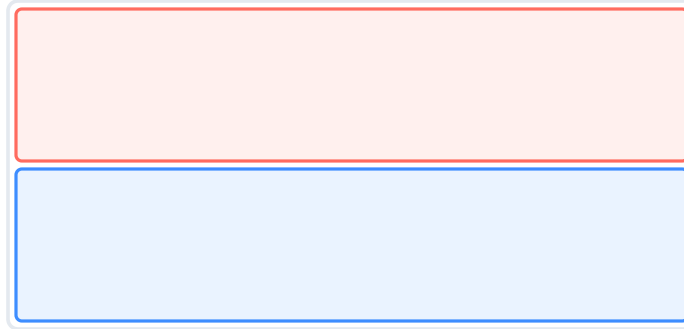
different times



time across, frequency up

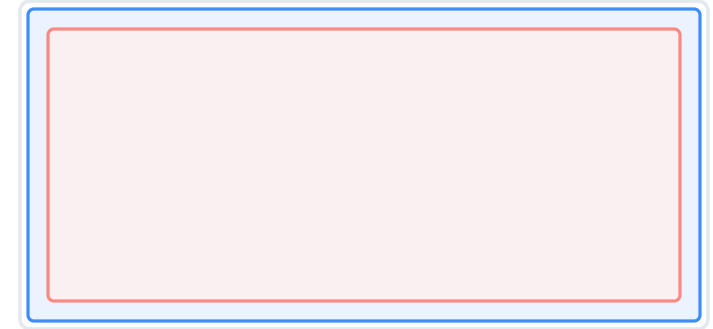
✓ a filter in time separates them

different frequencies



✓ a filter in frequency separates them

the same time and frequency



✗ no filter can separate them

— signal 1 — signal 2

so how do we separate the third one?

The first two are how every radio before this lecture shared a channel. The third one is what MIMO is for.

Two signals that overlap in both need something other than a filter.

1 Why MIMO

2 Diversity

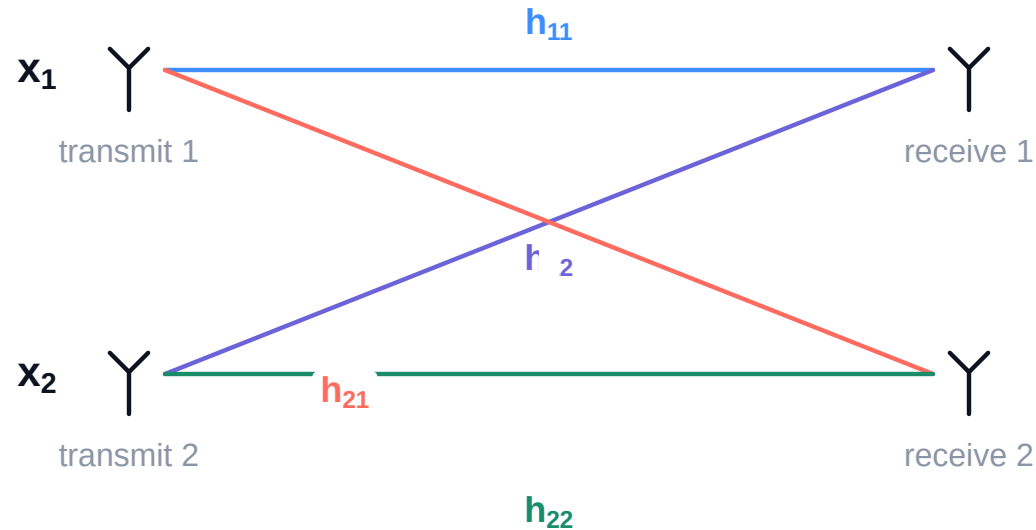
3 Multiplexing

4 Beamforming

5 Together

Send two different signals at once from two antennas

TWO TRANSMIT ANTENNAS, TWO RECEIVE ANTENNAS



$$y_1 = h_{11} x_1 + h_{12} x_2$$

$$y_2 = h_{21} x_1 + h_{22} x_2$$

two sums, four unknown gains, two unknown signals

Same time, same subcarrier, same modulation. The only thing that separates the two signals is where they left from.

Each receive antenna hears both transmissions, mixed in its own ratio.

1 Why MIMO

2 Diversity

3 Multiplexing

4 Beamforming

5 Together

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

THE SAME TWO LINES, WRITTEN ONCE

$$\begin{array}{c} \text{what arrives} \\ \left[\begin{array}{c} y_1 \\ y_2 \end{array} \right] \end{array} = \begin{array}{c} \text{the channel} \\ \left[\begin{array}{cc} h_{11} & h_{12} \\ h_{21} & h_{22} \end{array} \right] \end{array} \begin{array}{c} \text{what was sent} \\ \left[\begin{array}{c} x_1 \\ x_2 \end{array} \right] \end{array} + \begin{array}{c} \text{noise} \\ \left[\begin{array}{c} n_1 \\ n_2 \end{array} \right] \end{array}$$

one entry per antenna pair

h_{ij} is the gain from transmit j to receive i

one matrix per subcarrier

OFDM already made every subcarrier flat

complex, not real

each one is an amplitude and a phase

and it does not hold still

a new $\mathbf{H}$ every coherence time

$\mathbf{H}$ is not one number that got worse. It is a table of four numbers, and the four are what makes the problem solvable.

Everything in this lecture is a statement about the four numbers in $\mathbf{H}$.

1 Why MIMO

2 Diversity

3 Multiplexing

4 Beamforming

5 Together

Two unknowns and two equations

A CONCRETE CHANNEL

$$\begin{bmatrix} 0.8 & 0.3 \\ 0.4 & 0.9 \end{bmatrix}$$

H

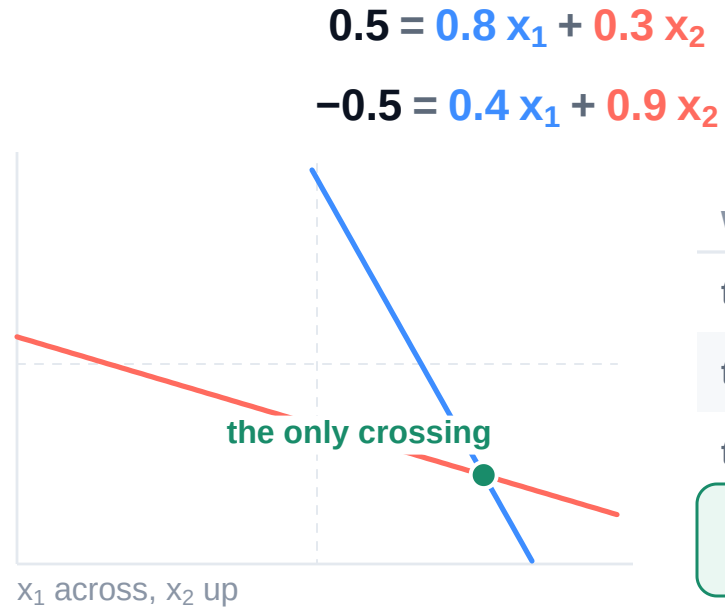
the transmitter sends

$$x_1 = +1 \quad x_2 = -1$$

the two antennas hear

$$y_1 = +0.5 \quad y_2 = -0.5$$

TWO EQUATIONS IN TWO UNKNOWNS



what is known	count
the two received numbers	2
the four entries of H	4
the two sent symbols	0
two equations, two unknowns, one answer	

Two lines that are not parallel cross in exactly one place, and that place is the pair of symbols that were sent.

Spatial multiplexing is a linear system, and it is solvable when the two lines are not parallel.

1 Why MIMO

2 Diversity

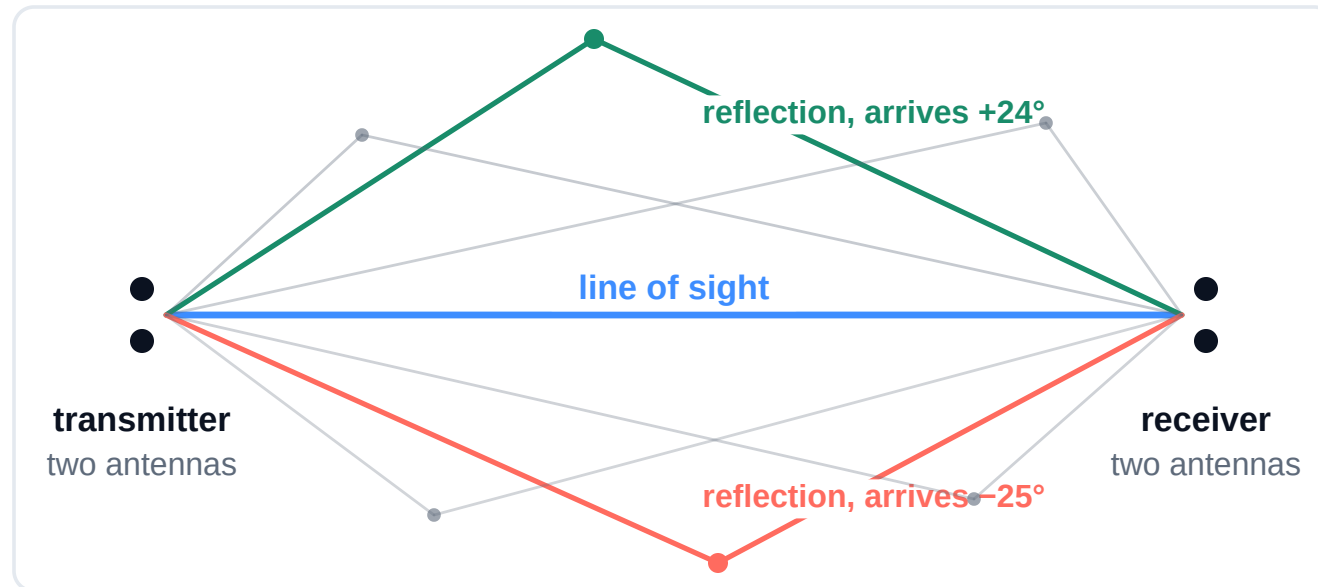
3 Multiplexing

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5 Together

A room offers one direct path and a few strong reflections

ONE ROOM, ONE TRANSMITTER, EVERY PATH IT OFFERS



- the direct path
strongest, one arrival angle
- two strong reflections
each one arrives from its own angle
- a crowd of weak scatter
too weak to carry anything on its own

Two streams need two paths the array can tell apart: different in the angle they leave by, or arrive by.

Drawn in plan view. Arrival angles are measured at the receive array; each path also leaves the transmitter at an angle of its own.

A second stream needs a second path, and the two have to be far apart in angle.

1 Why MIMO

2 Diversity

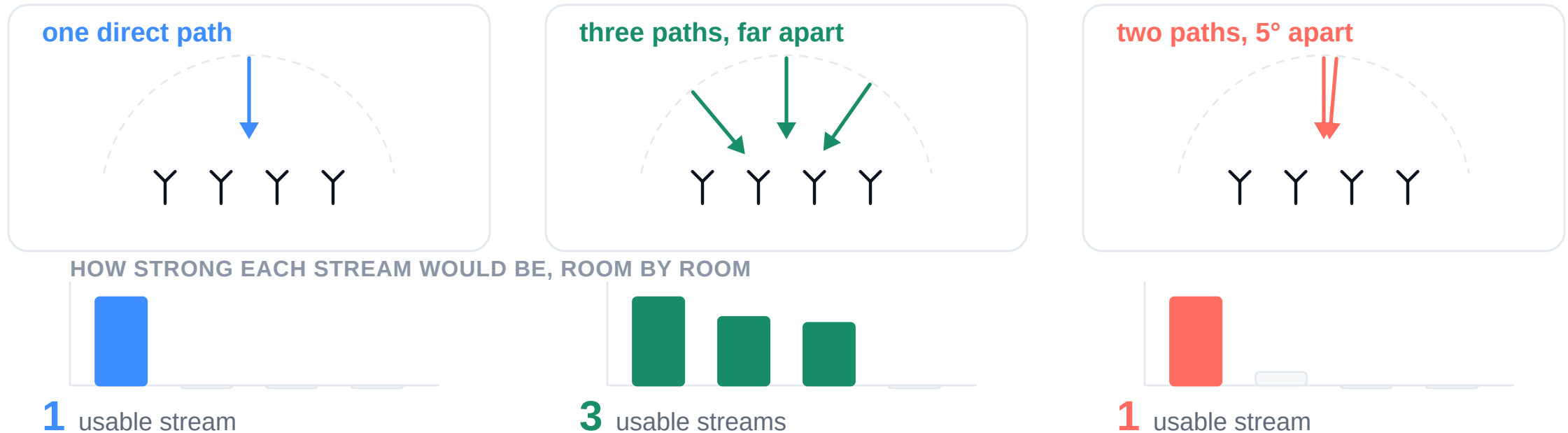
3 Multiplexing

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How many strong paths, how many streams

THE SAME FOUR-ELEMENT ARRAY, THREE ROOMS



streams $\approx$ min (antennas at each end, paths the array can tell apart)

Bars are the strength of each stream the room offers, strongest first. Two paths 5° apart give a second stream 16 dB down, which is no stream at all.

Count the paths the array can separate, and that is the number of streams.

1 Why MIMO

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Separating two streams can amplify the noise

A WELL-CONDITIONED CHANNEL

$$\begin{bmatrix} 0.8 & 0.3 \\ 0.4 & 0.9 \end{bmatrix}$$

singular values

1.21 and 0.50

condition number

2

noise gain on stream 2

$2 \times (6 \text{ dB})$

AN ILL-CONDITIONED CHANNEL

$$\begin{bmatrix} 0.80 & 0.79 \\ 0.40 & 0.41 \end{bmatrix}$$

singular values

1.26 and 0.01

condition number

133

noise gain on stream 2

$105 \times (40 \text{ dB})$

When the two columns of H point almost the same way,
separating the streams multiplies the noise.

The noise gain is one over the smallest singular value: the harder the two columns are to tell apart, the louder the answer.

The condition number of H decides whether two streams are worth sending at all.

1 Why MIMO

2 Diversity

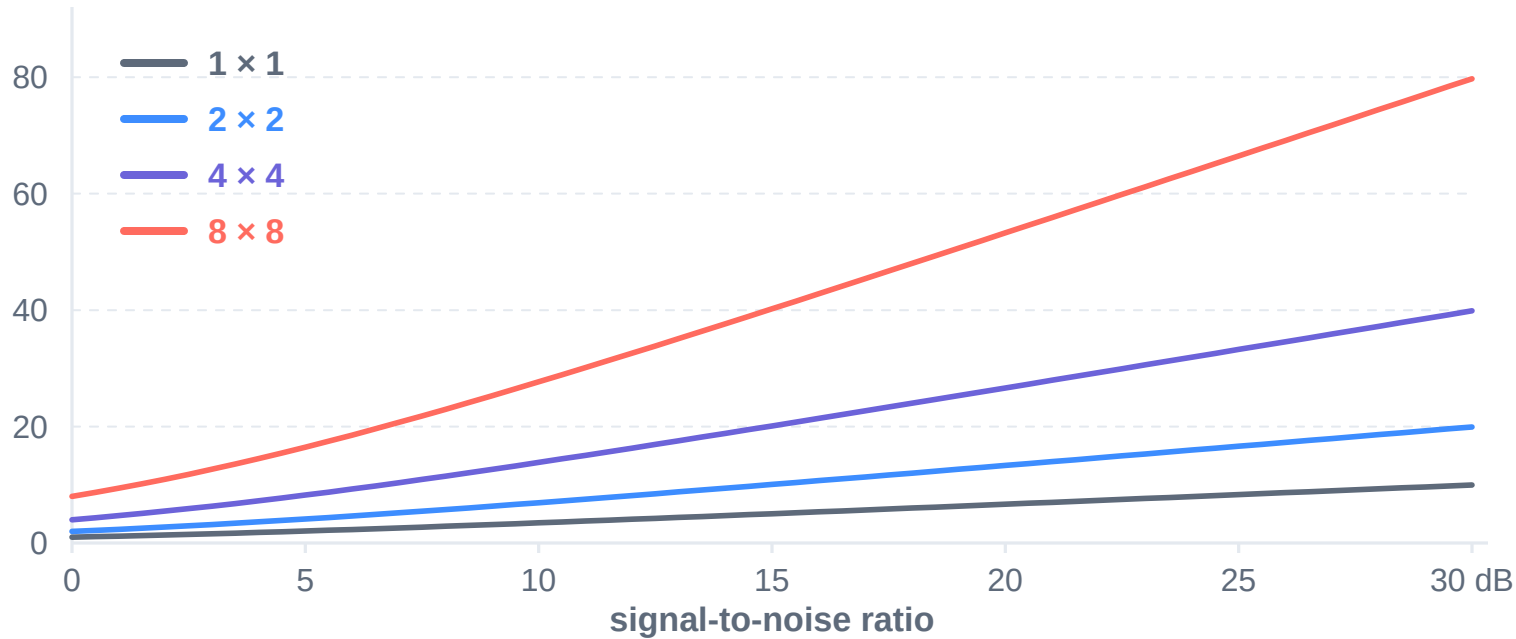
3 Multiplexing

4 Beamforming

5 Together

Capacity grows with the number of streams

BITS PER SECOND PER HERTZ, EQUAL PARALLEL PIPES



the slope is the
number of streams
power only moves the line up

Part 1 asked for a
multiplier, not an offset
this is the answer to it

Drawn for the ideal case of n equal, independent pipes. The slides before this one are about how far a real channel falls short of it.

Bandwidth and power move the line; antennas change its slope.

1 Why MIMO

2 Diversity

3 Multiplexing

4 Beamforming

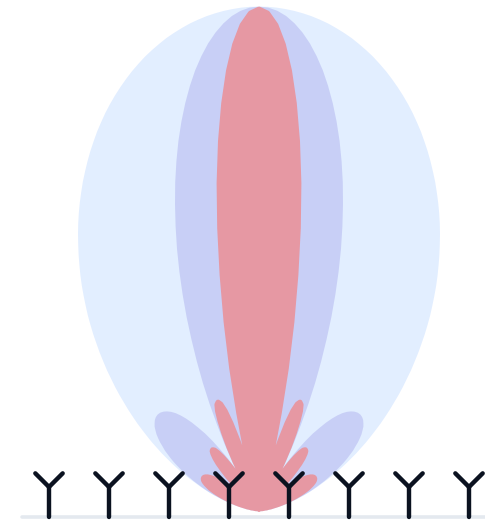
5 Together

Beamforming

How an array of antennas aims a signal

One antenna sends its power everywhere. Several antennas, each delayed by the right amount, send the same power in one chosen direction. This deck explains how, with one picture of two waves and no equations.

Yaxiong Xie



2, 4 and 8 antennas, aimed straight ahead

One antenna sends its power in every direction



receiver, 10 m away

1 in 4,000,000

of the transmitted power arrives

The rest goes in every other direction and is never received.

free space, 5 GHz, 10 m

The wave spreads out as a growing sphere. A receiving antenna a few centimetres across catches a very small part of that sphere.
At 10 m and 5 GHz the loss is 66 dB, before any wall is in the way.

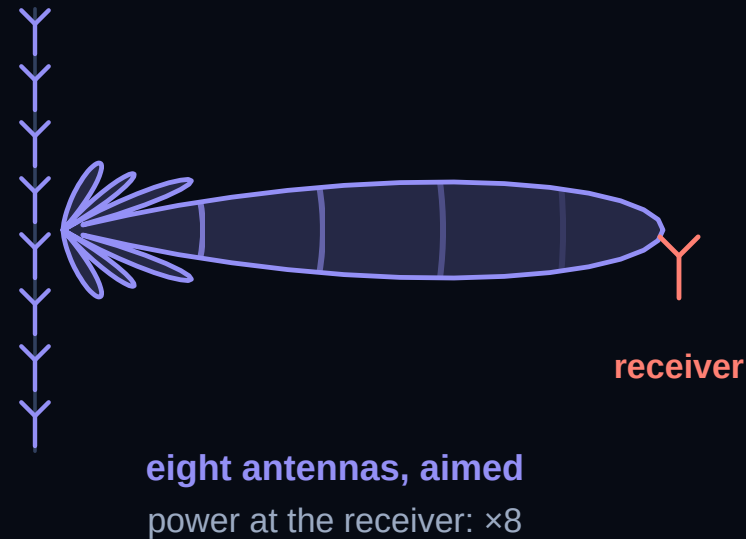
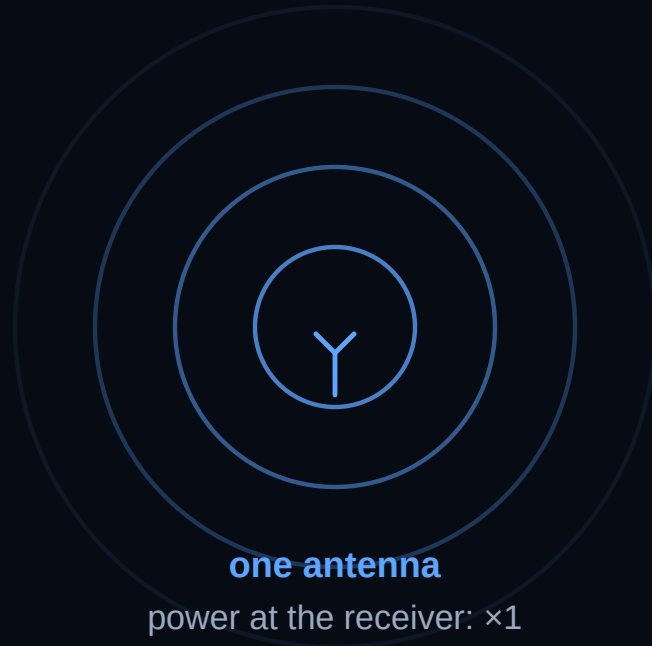
Most of the transmitted power goes where nobody is listening.

1 Why aim

2 How a beam forms

3 Two beams

Eight antennas can send the same power in one direction



$\times 8$
received power
same total power

Total transmit power is the same on both sides. On the right the eight copies of the signal add up in the direction of the receiver and cancel in the other directions.

Beamforming: many antennas, one signal, one direction.

1 Why aim

2 How a beam forms

3 Two beams

Beamforming is in Wi-Fi, in 5G and in satellite dishes

8

antennas

Wi-Fi 6 / 7 access point

aims a beam at each laptop

64

antennas

5G base station panel

3.5 GHz, one panel, 64 elements

1,280

elements

Starlink user terminal

steers to follow a satellite

Every 5G phone talks to a base station that beamforms. At 28 GHz a wavelength is 10.7 mm and antennas sit 5.4 mm apart, so an 8×8 panel of 64 antennas is about 4 cm across.

Every modern wireless system aims its signal instead of broadcasting it.

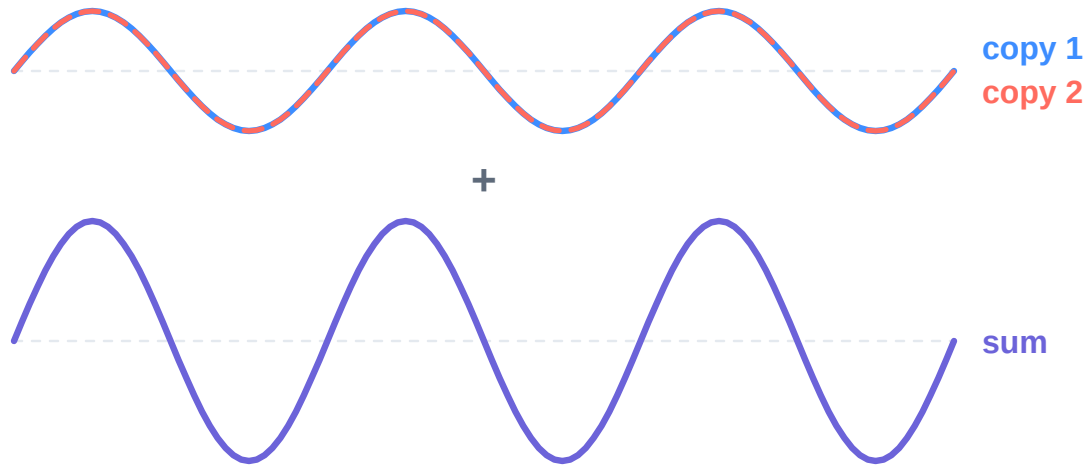
1 Why aim

2 How a beam forms

3 Two beams

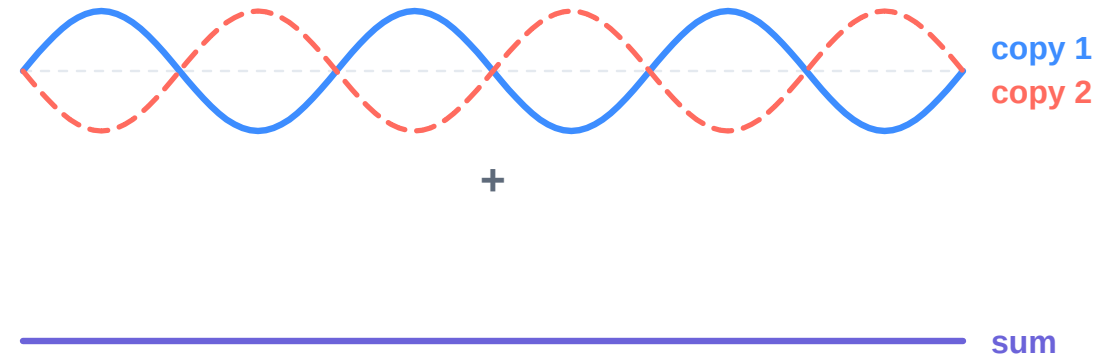
Two copies of a wave add up, or cancel, depending on timing

COPY 2 ARRIVES AT THE SAME TIME



the sum is twice as tall: $\times 2$ amplitude, $\times 4$ power

COPY 2 ARRIVES HALF A PERIOD LATE



shift 180°: the two copies cancel, sum $\times 0.0$

Same signal, two copies. If the second copy arrives at the same time, the peaks line up and the sum is twice as tall. If it arrives half a period late, peak meets trough and the sum is zero. On the right the delay of copy 2 changes continuously.

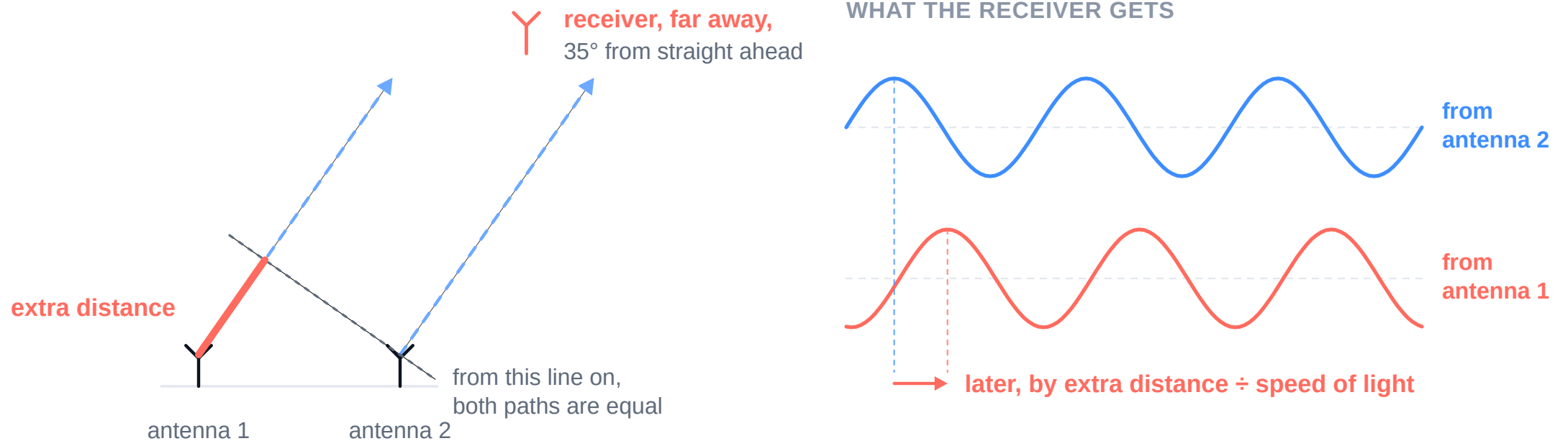
Everything in beamforming follows from this one picture.

1 Why aim

2 How a beam forms

3 Two beams

The copies from two antennas arrive at different times



Both antennas send the same signal. Seen from a receiver off to one side, antenna 1 is farther away, so its copy travels an extra distance and arrives later. The extra distance is the spacing times the sine of the angle.

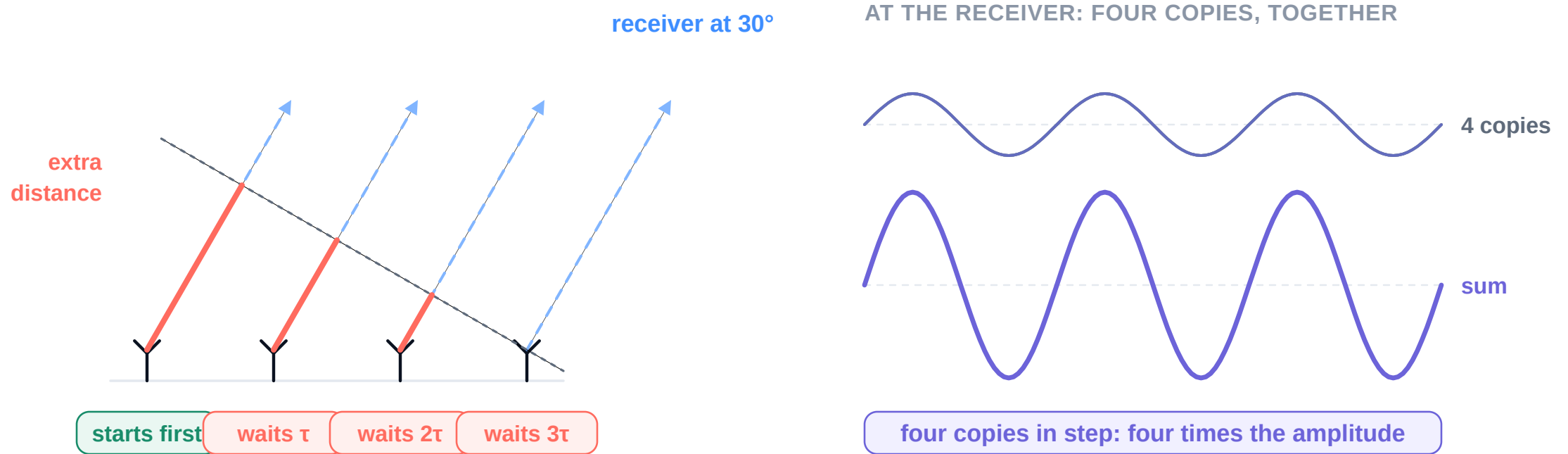
The angle of the receiver sets the delay between the copies.

1 Why aim

2 How a beam forms

3 Two beams

Delay each antenna a little, and the copies arrive together



Toward the receiver, the far antenna has the longest path, so it starts first. Each antenna closer to the receiver waits one extra delay τ , where $\tau = \text{extra distance} \div \text{speed of light}$. Antennas 6 cm apart, receiver at 30°: $\tau = 0.1 \text{ ns}$.

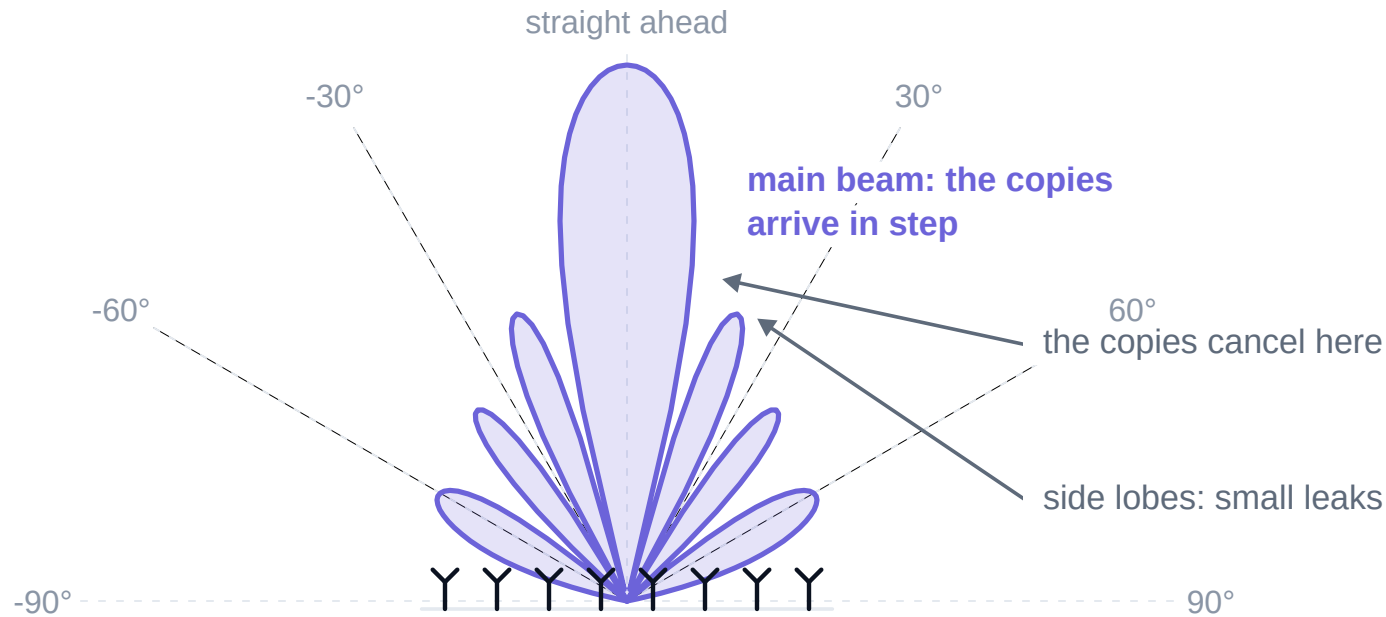
The right set of delays makes all copies arrive in step in one direction.

1 Why aim

2 How a beam forms

3 Two beams

In every other direction the copies cancel: this is the beam



beam width, half power **13°**

largest side lobe **5 % power**

gain straight ahead **×8**

Distance from the centre is the signal strength sent in that direction, in dB: the outer ring is the maximum, the centre is 1/1000 of it. Eight antennas.

The same delays that add the copies up straight ahead make them cancel a few degrees to either side. Plotting the strength in every direction gives this shape.

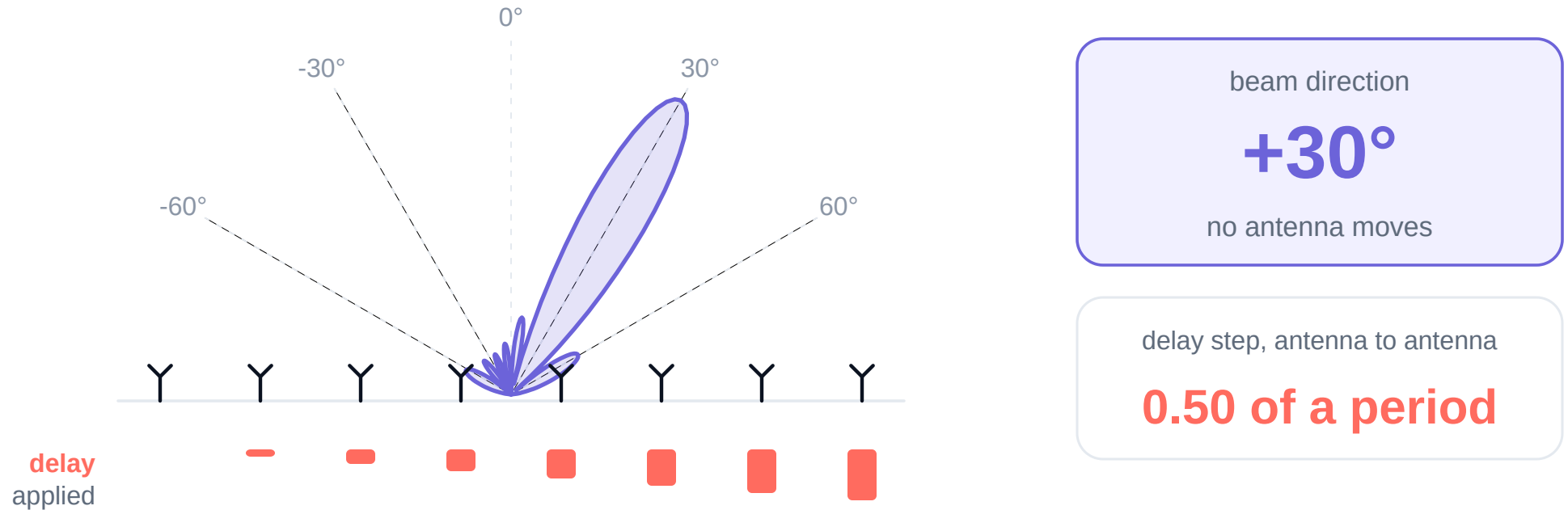
A beam is one direction where the copies add, surrounded by directions where they cancel.

1 Why aim

2 How a beam forms

3 Two beams

Change the delays and the beam turns; no antenna moves



Each red bar is the delay applied to one antenna, in periods of the wave. A larger step from antenna to antenna points the beam further from straight ahead. The change is made in the electronics; nothing mechanical moves.

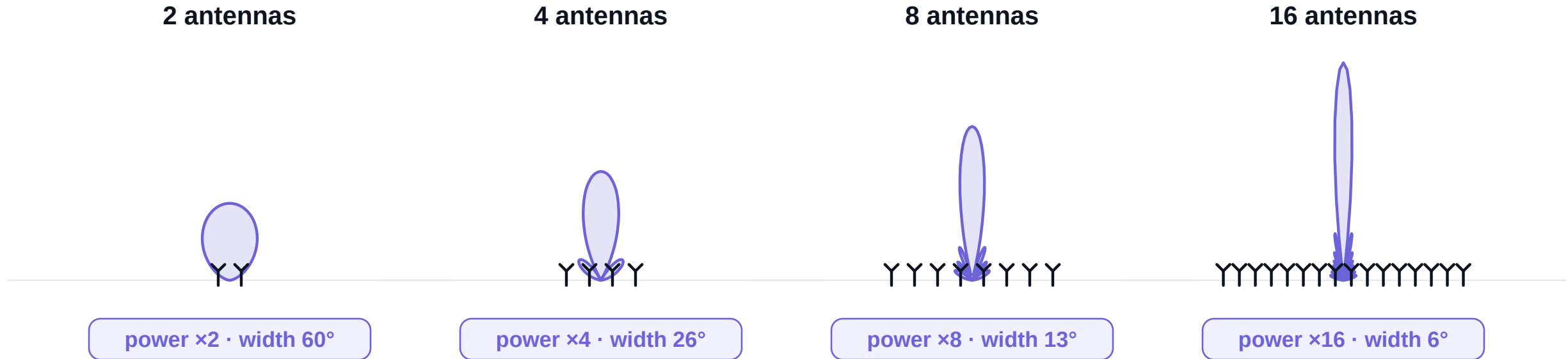
Steering a beam means changing eight numbers.

1 Why aim

2 How a beam forms

3 Two beams

More antennas make a stronger and narrower beam



Same total transmit power in every panel; the length of each lobe is the signal amplitude. Doubling the number of antennas doubles the power sent in the chosen direction and halves the beam width. 64 antennas: about 1.6° .

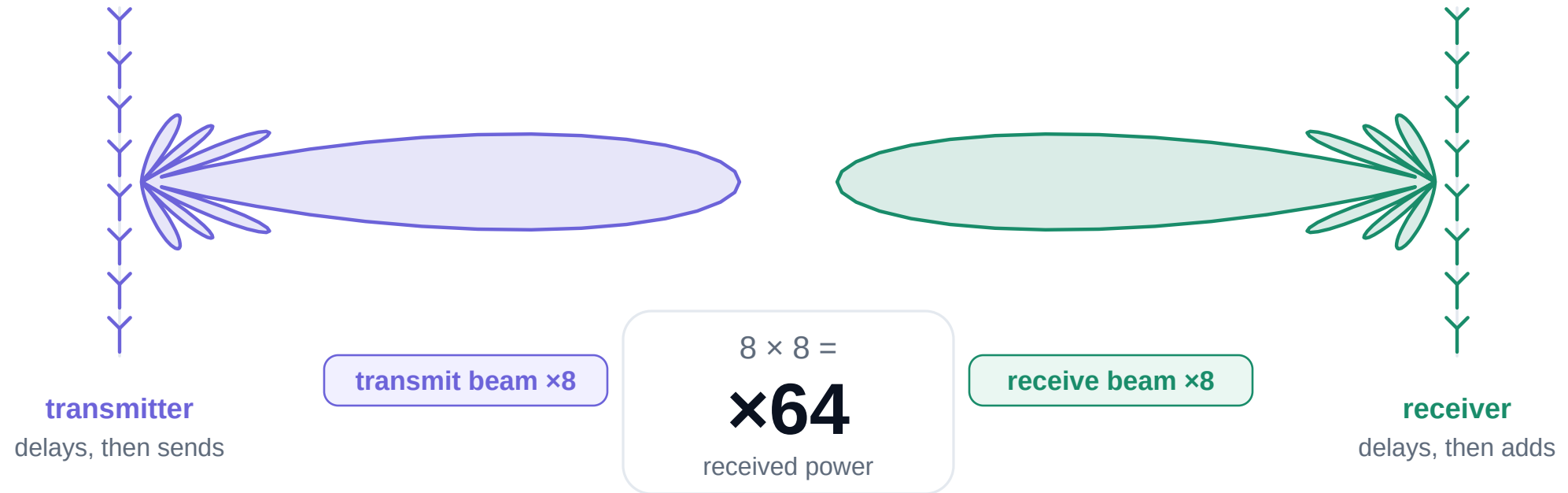
Gain and beam width both scale with the number of antennas.

1 Why aim

2 How a beam forms

3 Two beams

The receiver can form a beam too



A receiving array gets one copy of the signal per antenna. It delays each copy, then adds them. With the right delays the copies from one direction add up and copies from other directions cancel. The two gains multiply.

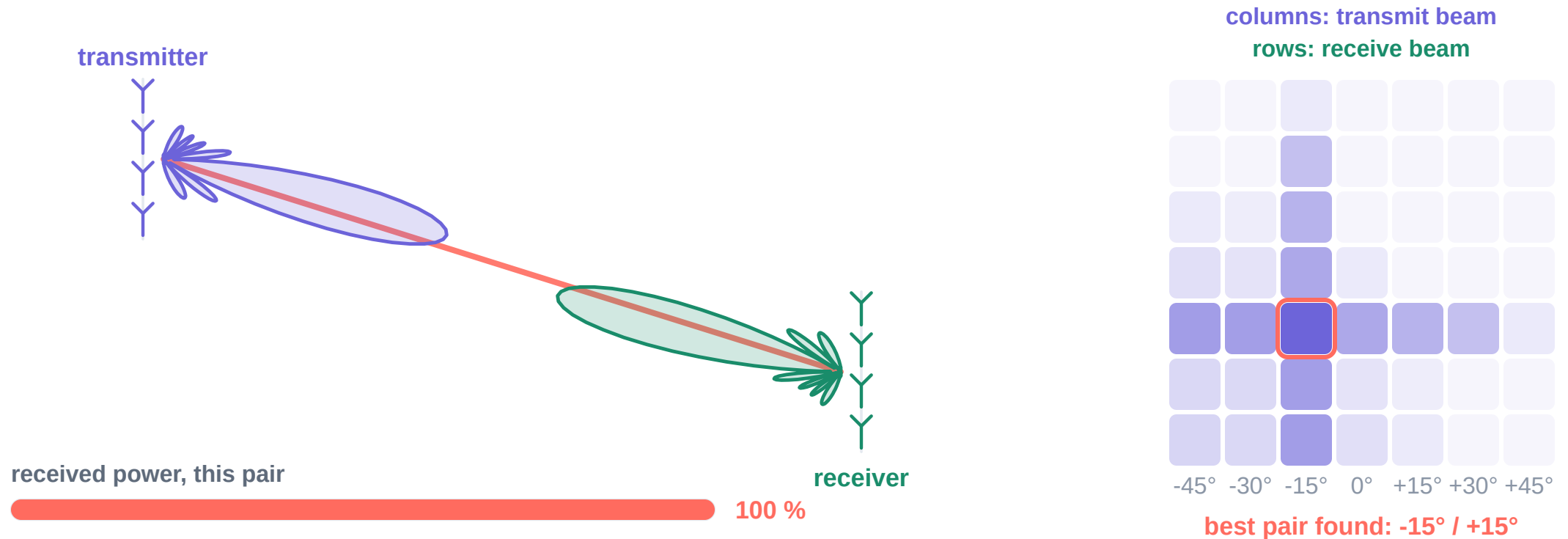
A link with a beam at each end gains at both ends: $8 \times 8 = 64$, or 18 dB.

1 Why aim

2 How a beam forms

3 Two beams

The two beams have to point at each other



Neither side knows where the other is. For every transmit beam, the receiver listens on each of its beams and records the power. Only one pair is strong. The grid fills in as the pairs are tried; the brightest cell is the answer.

Two narrow beams only help once they are aligned, and finding that pair takes a search.

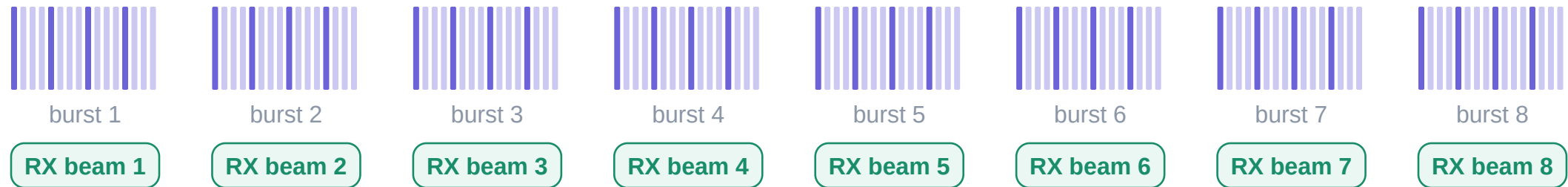
1 Why aim

2 How a beam forms

3 Two beams

Finding the pair: try every beam and keep the best

time → one burst of 64 transmit beams, every 20 ms



the phone listens with one receive beam per burst; after 8 bursts every pair has been tried

transmit beams	64	time per burst	20 ms	
receive beams	8	bursts needed	8	
pairs to try	$64 \times 8 = 512$		time to find the pair	160 ms

5G at 28 GHz: a base station sends up to 64 beams in turn in one burst and repeats the burst every 20 ms. The phone reports which beam was strongest. The 160 ms is an illustration for a phone that tries one receive beam per burst.

Before a single bit of data moves, the two ends spend time finding each other.

1 Why aim

2 How a beam forms

3 Two beams

A narrow beam is easy to lose

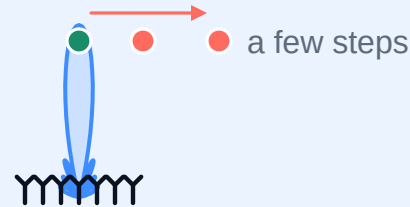
you turn the phone

The receive beam now points 20° away.
The received power drops to about 5 %.



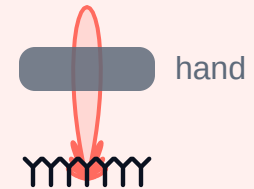
you walk

A 13° beam is 4.5 m wide at 20 m.
A few steps to the side and it misses.



your hand covers it

A hand or body in the path costs
about 30 dB at 28 GHz: $\times 1/1000$.



Beam tracking: keep testing the few beams next to the current one, and switch when a neighbour is stronger.

A full search is repeated only when the link is lost.

The gain of a narrow beam comes with a condition: both ends keep pointing at each other while people move. The 30 dB figure is the 3GPP self-blocking value for the user's own body.

The narrower the beam, the more often it has to be re-aimed.

1 Why aim

2 How a beam forms

3 Two beams

Beamforming in five sentences

- 1 One antenna sends power in every direction; almost none of it reaches the receiver.
- 2 Two copies of a wave add when they arrive together and cancel when they arrive half a period apart.
- 3 Delaying each antenna by the right amount makes the copies arrive together in one direction: a beam.
- 4 Changing the delays turns the beam. More antennas make it stronger and narrower.
- 5 The receiver can beamform too, and the two beams have to be found and kept pointed at each other.

Many antennas, one signal, chosen delays, one direction.

1 Why aim

2 How a beam forms

3 Two beams